



Advanced Time Series Analysis and Forecasting

Chapter 8: Advanced volatility modelling: realised measures, HAR and multivariate GARCH



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Acronyms used in this material (1/3)

- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model); in event studies: Abnormal Return
- ▣ ARCH: AutoRegressive Conditional Heteroskedasticity
- ▣ ARMA: AutoRegressive Moving Average (model)
- ▣ BEKK: Baba–Engle–Kraft–Kroner (multivariate GARCH model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BIC: Bayesian Information Criterion
- ▣ BNHLS: Barndorff–Nielsen–Hansen–Lunde–Shephard
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BV: Bipower Variation
- ▣ CAC: Cotation Assistée en Continu (CAC 40 index, Paris) — the CAC 40 index of the Paris bourse
- ▣ CCC: Constant Conditional Correlation (model)
- ▣ CLT: Central Limit Theorem
- ▣ DCC: Dynamic Conditional Correlation
- ▣ DCC-NL: DCC with Nonlinear shrinkage of the target (Engle–Ledoit–Wolf 2019)
- ▣ DM: Diebold–Mariano (test)
- ▣ EGARCH: Exponential GARCH
- ▣ EODHD: EOD Historical Data (market-data provider)
- ▣ ES: Expected Shortfall
- ▣ FDR: False Discovery Rate
- ▣ FRED: Federal Reserve Economic Data
- ▣ FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- ▣ FWER: Family-Wise Error Rate
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GARCH-MIDAS: GARCH with a Mixed Data Sampling long-run component (Engle–Ghysels–Sohn 2013)
- ▣ GARCH-X: GARCH with eXogenous variables

Acronyms used in this material (2/3)

- GJR: Glosten–Jagannathan–Runkle (asymmetric GARCH)
- GMM: Generalized Method of Moments
- GMV: Global Minimum-Variance (portfolio)
- HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- HAR: Heterogeneous AutoRegressive (model)
- HAR-CJ: HAR with Continuous and Jump components
- HARQ: HAR with realised Quarticity
- HEAVY: High-frEQUENCY-bAsed VolatilitY (models)
- IGARCH: Integrated GARCH
- INDPRO: Industrial production index (FRED series code)
- IQ: Integrated Quarticity
- LLM: Large Language Model
- MAE: Mean Absolute Error
- MCS: Model Confidence Set
- MFM: Modelling Financial Markets
- MIDAS: Mixed DAta Sampling (regression)
- ML: Machine Learning
- MSE: Mean Squared Error
- MSTR: MicroStrategy (Strategy Inc.), NASDAQ ticker
- OLS: Ordinary Least Squares
- OPG: Outer Product of the Gradient (scores)
- PPI: Producer Price Index
- QLIKE: Quasi-LIKElihood loss
- QML: Quasi-Maximum Likelihood
- QMLE: Quasi-Maximum Likelihood Estimator
- QV: Quadratic Variation

Acronyms used in this material (3/3)

- ▣ RGARCH: Realized GARCH
- ▣ RK: Realised Kernel
- ▣ RMSE: Root Mean Squared Error
- ▣ RQ: Realised Quarticity
- ▣ RV: Realised Variance
- ▣ SV: Stochastic Volatility
- ▣ TQ: Tripower Quarticity
- ▣ TSA: Time Series Analysis
- ▣ TSLA: Tesla Inc., NASDAQ ticker
- ▣ TSRV: Two-Scales Realised Variance
- ▣ US: United States
- ▣ UTC: Coordinated Universal Time (Temps Universel Coordonné)
- ▣ VEC: VEC multivariate GARCH (half-vectorised covariance)
- ▣ VIX: CBOE Volatility Index
- ▣ VR: Variance Ratio
- ▣ WPSFD49207: PPI by commodity: final demand, finished goods (FRED series code)

Today's question and route

- ▣ **Question:** how precisely can we measure, model and forecast a variance that is never observed?
 - ▶ and which of our tools still work when the data are noisy, jumpy or high-dimensional?
 - ▶ two routes to volatility: a parametric filter of daily returns (GARCH) and a nonparametric measurement from intraday prices (realised measures); the frontier combines them
- ▣ **Route** of the chapter
 - ▶ quasi-maximum likelihood for GARCH and robust inference; long-run components: component GARCH and GARCH-MIDAS with macroeconomic drivers
 - ▶ realised measures: asymptotics of realised variance, microstructure noise, realised kernels, bipower variation and jump tests
 - ▶ forecasting with realised measures: HAR, HARQ, Realized GARCH, HEAVY; robust loss functions
 - ▶ multivariate: BEKK, DCC and cDCC, the curse of dimensionality and DCC-NL, realised covariance
- ▣ We build on [TSA, Chapter 5](#) (GARCH, GJR, EGARCH, news impact, QLIKE) and [TSA, Chapter 14](#) (DCC basics)
 - ▶ MFM, Chapters 5–9 apply these tools to market risk; Seminar 8 comes before this lecture

Learning outcomes

- ▣ State the conditions under which Gaussian QML for GARCH is consistent and asymptotically normal, and compute Bollerslev–Wooldridge standard errors
- ▣ Separate short-run from long-run volatility with component GARCH and GARCH-MIDAS, and judge whether a macroeconomic driver matters
- ▣ Derive the limit theory of realised variance, explain the effect of noise and jumps, and choose among RV, realised kernels and bipower variation
- ▣ Forecast volatility with HAR, HARQ and Realized GARCH, and compare forecasts with losses that are robust to a noisy proxy
- ▣ Estimate DCC and cDCC models, explain why large covariance matrices need shrinkage, and measure correlation from intraday data

Reading, data and tools

- Backbone: Francq and Zakoïan (2019); Andersen, Bollerslev, Diebold and Labys (2003); Barndorff-Nielsen and Shephard (2002); Barndorff-Nielsen, Hansen, Lunde and Shephard (2008)
 - ▶ forecasting and the multivariate case: Corsi (2009); Hansen, Huang and Shek (2012); Engle (2002); Patton (2011)
 - ▶ case studies: Engle, Ghysels and Sohn (2013) (GARCH-MIDAS), Bollerslev, Patton and Quaadvlieg (2016) (HARQ), Hansen, Huang and Shek (2012) (Realized GARCH), Engle, Ledoit and Wolf (2019) (DCC-NL)
- Python Quantlets of this chapter: [Quantlets/Ch_08](#)
 - ▶ QML with sandwich standard errors, component GARCH, GARCH-MIDAS, realised kernels, jump tests, HAR/HARQ, Realized GARCH, DCC/cDCC and nonlinear shrinkage written out in `numpy`
- Lecture notebook: [open in Google Colab](#)

Data used in this chapter

Series	Source	Use
Daily realised measures of six equity indices (5-minute RV, BV, realised kernel, open-to-close return), 2000–2022	Heber, Lunde, Shephard and Sheppard (2009) , v0.3	HAR, RGARCH
Bitcoin and Ether: one-minute prices 2018–2026, one-second prices August 2026	Binance public data (data.binance.vision); measures computed by us	noise, jumps, HARQ
S&P 500, DAX, BET, Bitcoin, 14 US stocks and 3 equity ETFs, daily	EODHD	QML, MIDAS, DCC
EUR/RON reference rate	BNR	QML
US industrial production, PPI finished goods (monthly)	FRED (INDPRO, WPSFD49207)	GARCH-MIDAS

- The Oxford-Man library ended in February 2022 (archived copy, cited as its terms require); Binance one-minute data extend the realised measures to 18 September 2026; a calibrated simulation gives the theory a known truth
- RV: realised variance; BV: bipower variation; PPI: producer price index

From the trading floor to the tick



New York Stock Exchange, 1908

Photo: Helen D. Van Eaton, *Leslie's Monthly Magazine* (1908);
public domain; [Wikimedia Commons](#)

- 1982–1986: ARCH and GARCH Engle (1982), Bollerslev (1986): volatility as a filtered conditional variance of daily returns
- 1992–2004: QML theory Bollerslev and Wooldridge (1992), Lee and Hansen (1994), Lumsdaine (1996), Francq and Zakoïan (2004); components Engle and Lee (1999); DCC Engle (2002)
- 1998–2008: realised variance Andersen and Bollerslev (1998), Andersen, Bollerslev, Diebold and Labys (2003), Barndorff-Nielsen and Shephard (2002); bipower variation Barndorff-Nielsen and Shephard (2004); noise and kernels Zhang, Mykland and Aït-Sahalia (2005), Barndorff-Nielsen, Hansen, Lunde and Shephard (2008)
- 2009–2019: HAR Corsi (2009), Realized GARCH Hansen, Huang and Shek (2012), GARCH-MIDAS Engle, Ghysels and Sohn (2013), HARQ Bollerslev, Patton and Quaedvlieg (2016), DCC-NL Engle, Ledoit and Wolf (2019)
- Today: high-frequency data are cheap; the questions are statistical (noise, jumps, dimension, evaluation)

From TSA to this chapter

- Known ([TSA, Chapter 5](#)): GARCH(1,1), GJR, EGARCH, the news impact curve, ML with Gaussian and Student- t errors, QLIKE; ([TSA, Chapter 14](#)): CCC and DCC
- New here
 - ▶ why Gaussian ML still works when returns are not Gaussian, and which standard errors are then valid
 - ▶ volatility with a slow, macro-driven component; volatility measured from intraday prices, with its own sampling error
 - ▶ models that use realised measures; evaluation against a noisy target; covariance matrices with many assets
- Stochastic volatility (a latent log-variance with its own shock): [Chapter 6](#); Markov-switching GARCH: [Chapter 7](#); VaR and ES backtesting: [Chapter 9](#); rough volatility: [Chapter 10](#)

The Gaussian quasi-likelihood (1/2): the model

- The return shock is a volatility times a standardised shock; the variance follows a GARCH(1,1) recursion

$$\varepsilon_t = \sigma_t(\theta_0) \eta_t, \quad \sigma_t^2(\theta) = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2(\theta)$$

- Notation

- ▶ ε_t : the demeaned return of day t ; $\sigma_t^2(\theta)$: its variance conditional on the past $\mathcal{F}_{t-1}$ (the information up to day $t-1$)
 - ▶ η_t : i.i.d. standardised shock, $\mathbb{E}\eta_t = 0$, $\mathbb{E}\eta_t^2 = 1$; its law (Normal, Student- t , skewed) is unknown
 - ▶ $\theta = (\omega, \alpha, \beta)$, true value θ_0 : $\omega > 0$ sets the level, $\alpha \geq 0$ the reaction to yesterday's squared shock, $\beta \geq 0$ the persistence of yesterday's variance
- The recursion starts from an arbitrary σ_1^2 ; the effect of the start vanishes geometrically when $\beta < 1$

The Gaussian quasi-likelihood (2/2): estimator and score

- QMLE: maximise the Gaussian log-likelihood, used as an estimating equation even if η_t is not Gaussian

$$\hat{\theta} = \arg \max_{\theta} \sum_{t=1}^T \ell_t(\theta), \quad \ell_t(\theta) = -\frac{1}{2} \left(\ln \sigma_t^2(\theta) + \frac{\varepsilon_t^2}{\sigma_t^2(\theta)} \right)$$

► ℓ_t : the Gaussian log-density of day t without its constant; T : number of days; $\hat{\theta}$: the estimate

- The score, the gradient of ℓ_t , has conditional mean zero at the true value

$$s_t(\theta) = \frac{\partial \ell_t}{\partial \theta} = \frac{1}{2} \left(\frac{\varepsilon_t^2}{\sigma_t^2} - 1 \right) \frac{1}{\sigma_t^2} \frac{\partial \sigma_t^2}{\partial \theta}$$

► at θ_0 , $\varepsilon_t^2 / \sigma_t^2 = \eta_t^2$, hence $\mathbb{E}(s_t | \mathcal{F}_{t-1}) = 0$ because $\mathbb{E}\eta_t^2 = 1$

► only the conditional variance must be right; the shape of the law of η_t is never used

- The score is a martingale difference (conditional mean zero given the past)

► a CLT for martingales gives asymptotic normality without independence of ε_t

Consistency and asymptotic normality (1/2): stationarity and consistency

- GARCH(1,1) is strictly stationary if and only if the log of the daily variance multiplier has a negative mean

$$\mathbb{E} \ln(\alpha_0 \eta_t^2 + \beta_0) < 0$$

- ▶ $\alpha_0 \eta_t^2 + \beta_0$: the factor by which a shock to the variance is carried from one day to the next; α_0, β_0 : true values
- ▶ by Jensen's inequality, weaker than $\alpha_0 + \beta_0 < 1$ (finite variance): IGARCH ($\alpha_0 + \beta_0 = 1$) is strictly stationary, and $\mathbb{E} \varepsilon_t^2$ may be infinite
- ▶ Lee and Hansen (1994), Lumsdaine (1996): GARCH(1,1), including IGARCH; Berkes, Horváth and Kokoszka (2003), Francq and Zakoïan (2004): GARCH(p, q) under strict stationarity
- Consistency of $\hat{\theta}$ Francq and Zakoïan (2004)
 - ▶ conditions: strict stationarity, θ_0 in a compact parameter set, identifiability (η_t^2 is not a constant)
 - ▶ no moment of ε_t is needed
- Even explosive GARCH ($\mathbb{E} \ln(\alpha_0 \eta_t^2 + \beta_0) > 0$): $(\hat{\alpha}, \hat{\beta})$ remain consistent and asymptotically normal, ω is not identified Jensen and Rahbek (2004)

Consistency and asymptotic normality (2/2): the limit law

- With θ_0 interior to the parameter set and a finite fourth moment of η_t , the estimator is asymptotically normal

$$\sqrt{T}(\hat{\theta} - \theta_0) \rightarrow N(0, (\kappa_\eta - 1)J^{-1}), \quad J = \mathbb{E} \left[\frac{1}{\sigma_t^4} \frac{\partial \sigma_t^2}{\partial \theta} \frac{\partial \sigma_t^2}{\partial \theta'} \right]$$

- ▶ $\kappa_\eta = \mathbb{E}\eta_t^4$: kurtosis of the shock (3 for the Normal distribution); $\kappa_\eta - 1 = \text{Var}(\eta_t^2)$
- ▶ J : a 3×3 matrix, the mean outer product of the relative sensitivities $\sigma_t^{-2} \partial \sigma_t^2 / \partial \theta$; $\rightarrow$: convergence in distribution as $T \rightarrow \infty$
- ▶ reading: the sampling variance of $\hat{\theta}$ grows linearly with the kurtosis of the shocks
- Boundary: if $\alpha_0 = 0$, θ_0 is not interior and the limit is the projection of a normal vector on a cone
 - ▶ this is why [TSA, Chapter 5](#) tests ARCH effects one-sided

The sandwich and the Bollerslev–Wooldridge standard errors (1/2)

- General QML result: the asymptotic variance is a “sandwich” of two matrices

$$\sqrt{T}(\hat{\theta} - \theta_0) \rightarrow N(0, A^{-1} B A^{-1}), \quad A = -\mathbb{E} \frac{\partial^2 \ell_t}{\partial \theta \partial \theta'}, \quad B = \mathbb{E} s_t s_t'$$

- ▶ A : the expected negative Hessian (curvature of the log-likelihood); B : the variance of the score s_t
- ▶ if the likelihood is the true one, $A = B$ (information equality) and the variance reduces to A^{-1} ; under QML, $A \neq B$
- For GARCH both matrices are proportional to J (▶ [Appendix](#))

$$A = \frac{1}{2} J, \quad B = \frac{\kappa_\eta - 1}{4} J, \quad A^{-1} B A^{-1} = (\kappa_\eta - 1) J^{-1}$$

- ▶ the sandwich recovers the limit law of the previous slide, for any law of η_t with $\kappa_\eta < \infty$

The sandwich and the Bollerslev–Wooldridge standard errors (2/2)

□ Three ways to compute standard errors (s.e.)

- ▶ Hessian only: $A^{-1} = 2J^{-1}$, correct only if $\kappa_\eta = 3$; too small by the factor $\sqrt{(\kappa_\eta - 1)/2}$ when $\kappa_\eta > 3$
- ▶ outer product of the scores (OPG): $B^{-1} = \frac{4}{\kappa_\eta - 1} J^{-1}$, even smaller when $\kappa_\eta > 3$
- ▶ sandwich Bollerslev and Wooldridge (1992): $\hat{A}^{-1} \hat{B} \hat{A}^{-1}$, valid whatever the law of η_t (with $\kappa_\eta < \infty$)

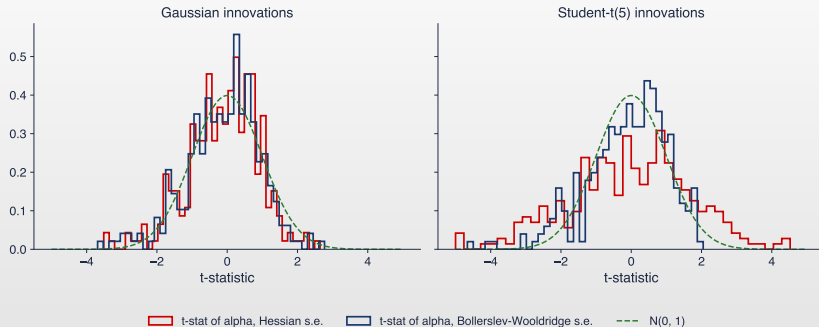
□ Estimates used by Bollerslev and Wooldridge

- ▶ $\hat{A}$: the observed Hessian of $-\frac{1}{T} \sum_t \ell_t$ at $\hat{\theta}$; $\hat{B} = \frac{1}{T} \sum_t \hat{s}_t \hat{s}_t'$, with $\hat{s}_t = s_t(\hat{\theta})$ the estimated score

□ Robust Wald and score tests use the sandwich

- ▶ the quasi-likelihood-ratio (quasi-LR) statistic is no longer χ^2 , but a weighted sum of χ_1^2 variables (chi-square with one degree of freedom)

Hessian against sandwich: a Monte Carlo

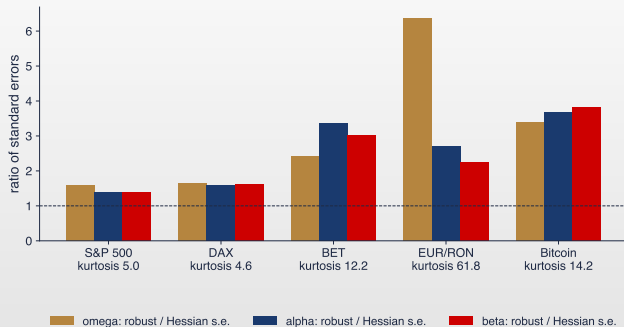


- GARCH(1,1) with $(\omega, \alpha, \beta) = (0.05, 0.08, 0.90)$, $T = 2000$, 300 replications; Gaussian QML in each; histogram of $(\hat{\alpha} - \alpha_0)/\hat{s.e.}$

Interpreting the Monte Carlo

- ▣ Gaussian η : coverage of nominal 95% intervals for α is 95.0% (Hessian) and 94.3% (sandwich): both fine
- ▣ Student- $t(5)$, $\kappa_\eta = 9$: Hessian intervals cover 73.0% for α and 74.7% for β ; sandwich intervals 91.7% and 92.3%
- ▣ The theory predicts the factor $\sqrt{(9-1)/2} = 2$ between the two standard errors; the point estimates are the same
- ▣ With fat tails, a Hessian-based t -test of α rejects far too often: report sandwich standard errors by default

Robust and naive standard errors in five markets



- Gaussian QML of GARCH(1,1), daily returns 2010–2026; bars: sandwich s.e. divided by Hessian s.e.; labels: kurtosis $\hat{\kappa}_\eta$ of the standardised residuals

Interpreting the five markets

- ▣ S&P 500: $\hat{\alpha} = 0.154$ with s.e. 0.013 (Hessian) against 0.018 (sandwich); $\hat{\kappa}_\eta = 5.0$, theoretical factor 1.42
- ▣ BET: $\hat{\kappa}_\eta = 12.2$, the sandwich s.e. of $\hat{\alpha}$ is 3.36 times larger; EUR/RON (managed float): 2.71 times, $\hat{\kappa}_\eta = 61.8$
- ▣ The ratios track $\sqrt{(\hat{\kappa}_\eta - 1)/2}$ only roughly: in the data η_t is not i.i.d., and the sandwich does not need it to be
- ▣ Student- t ML raises the log-likelihood by 124.4 (S&P 500) and 233.0 (BET), but is consistent only if the t shape is right; QML needs only the variance equation

Recap: QML for GARCH

- Gaussian QML is consistent if the variance equation is right, whatever the shape of the innovations
- Strict stationarity, not finite variance, is what the theory needs; IGARCH and even explosive GARCH are covered
- With fat tails, Hessian standard errors are too small by $\sqrt{(\kappa_\eta - 1)/2}$; use Bollerslev–Wooldridge

GARCH with exogenous variables

- GARCH-X: yesterday's value of an observed variable enters the variance equation

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 + \pi' x_{t-1}$$

- ▶ x_{t-1} : vector of exogenous variables known at the end of day $t - 1$; π : their loadings ($\pi'x$: the weighted sum)
- ▶ positivity of σ_t^2 : $x_{t-1} \geq 0$ and $\pi \geq 0$, or a specification in logs
- Candidates for x_t : a realised measure (leads to HEAVY and Realized GARCH below), the VIX, a macro or policy-uncertainty index, trading volume
 - ▶ QML theory carries over if x_t is stationary and ergodic; its own dynamics need not be modelled for one-step forecasts
 - ▶ multi-step forecasts need a model for x_t : this is exactly what Realized GARCH and HEAVY add
- A slow variable (a monthly macro series) cannot enter a daily recursion at its own frequency: one needs a component structure (next slides)

The component GARCH of Engle and Lee (1/2): the model

- Engle and Lee (1999): the variance is a slowly moving long-run level q_t plus a transitory deviation from it

$$\sigma_t^2 = q_t + \alpha(\varepsilon_{t-1}^2 - q_{t-1}) + \beta(\sigma_{t-1}^2 - q_{t-1})$$

$$q_t = \omega + \rho q_{t-1} + \phi(\varepsilon_{t-1}^2 - \sigma_{t-1}^2)$$

- ▶ q_t : long-run (permanent) component; ρ : its persistence, close to 1; ω : its intercept
 - ▶ $\sigma_t^2 - q_t$: transitory component; α, β : its reaction and persistence, with $\alpha + \beta < \rho$
 - ▶ $\varepsilon_{t-1}^2 - \sigma_{t-1}^2$: the variance surprise of yesterday (mean zero); ϕ : how much of it moves the long-run level
- Both components are driven by the same shock: the model is a restricted GARCH(2,2)

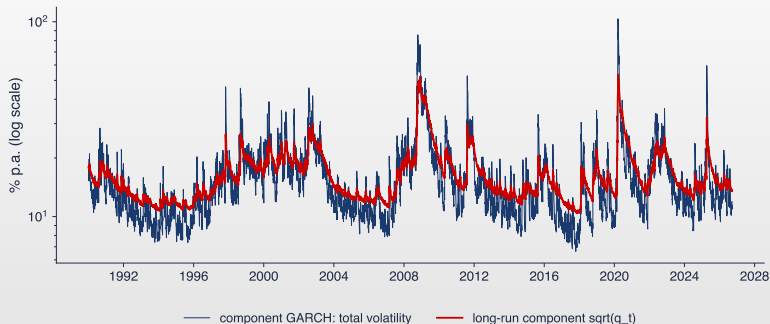
The component GARCH of Engle and Lee (2/2): two speeds

- Half-life: the number of days after which half of a shock to a component has died out

$$HL_q = \frac{\ln(1/2)}{\ln \rho}, \quad HL_{\sigma-q} = \frac{\ln(1/2)}{\ln(\alpha + \beta)}$$

- ▶ HL_q : half-life of the long-run component; $HL_{\sigma-q}$: half-life of the transitory component
- ▶ a persistence $\rho = 0.99$ gives about 69 days; $\alpha + \beta = 0.9$ gives about 6.6 days
- ▶ GARCH(1,1) forces a single half-life, $\ln 0.5 / \ln(\alpha + \beta)$
- Two exponentials approximate a slowly decaying (hyperbolic) autocorrelation of squared returns over a finite range of lags
 - ▶ long memory proper: [Chapter 10](#)

Component GARCH for the S&P 500



□ Daily returns since 1990, $T = 9240$; Gaussian QML; annualised total volatility $\sqrt{252\sigma_t^2}$ and long-run component $\sqrt{252q_t}$

Interpreting the component model

- ▣ $\hat{\rho} = 0.995$, $\hat{\phi} = 0.026$; transitory part $\hat{\alpha} + \hat{\beta} = 0.950$: half-lives 150 days (long run) and 13.6 days (short run)
- ▣ GARCH(1,1) on the same data: persistence 0.983, one half-life of 41 days, a compromise between the two
- ▣ Quasi-LR statistic against GARCH(1,1): 44.1; under the null $\phi = 0$ the parameter ρ is not identified (Davies problem), so the χ^2_2 reference is only indicative
- ▣ After 2008, 2020 and 2025 the long-run level stays high for months while daily volatility has already fallen: this matters for horizons beyond a few weeks

GARCH-MIDAS (1/3): a macro-driven long-run component



Eric Ghysels, 2019

Photo: Eghysels (2019); CC BY-SA 4.0;

Wikimedia Commons

- Engle, Ghysels and Sohn (2013): the daily variance is the product of a monthly level τ_t and a daily factor $g_{i,t}$

$$r_{i,t} = \mu + \sqrt{\tau_t g_{i,t}} \eta_{i,t}$$

- ▶ $r_{i,t}$: return of day i of month t ; μ : mean return; $\eta_{i,t}$: i.i.d. shock with mean 0 and variance 1

- Short run: a GARCH(1,1) with mean 1, re-scaled by τ_t

$$g_{i,t} = (1 - \alpha - \beta) + \alpha \frac{(r_{i-1,t} - \mu)^2}{\tau_t} + \beta g_{i-1,t}$$

- ▶ α, β : reaction and persistence of the daily component, as in GARCH(1,1)

GARCH-MIDAS (2/3): the long-run component

- Long run: MIDAS (mixed-data sampling) regression of $\ln \tau_t$ on K lags of a monthly variable

$$\ln \tau_t = m + \theta \sum_{k=1}^K \varphi_k(w) X_{t-k}, \quad \varphi_k(w) \propto (1 - k/K)^{w-1}$$

- ▶ X_{t-k} : the monthly driver k months ago; m : intercept; θ : effect of the driver (sign: pro- or countercyclical)
- ▶ $\varphi_k(w)$: beta lag weights, normalised to sum to 1 and restricted to decay [Ghysels, Santa-Clara and Valkanov \(2006\)](#); $w \geq 1$: the larger w , the faster the decay; $w = 1$: equal weights
- Choices of X
 - ▶ realised variance of past months, in levels: $\tau_t = m + \theta \sum_k \varphi_k(w) \text{RV}_{t-k}$, with RV_{t-k} the sum of squared daily returns of month $t - k$
 - ▶ macro data: industrial production growth, PPI inflation

GARCH-MIDAS (3/3): identification, estimation and the variance ratio

- τ_t is constant within the month and $\mathbb{E}g_{i,t} = 1$: the split is identified by the different frequencies, without a filter for a latent variable
 - ▶ Gaussian QML of all parameters $(\mu, \alpha, \beta, m, \theta, w)$ in one step; sandwich standard errors
- Variance ratio: the share of the variation of log volatility explained by the long-run component

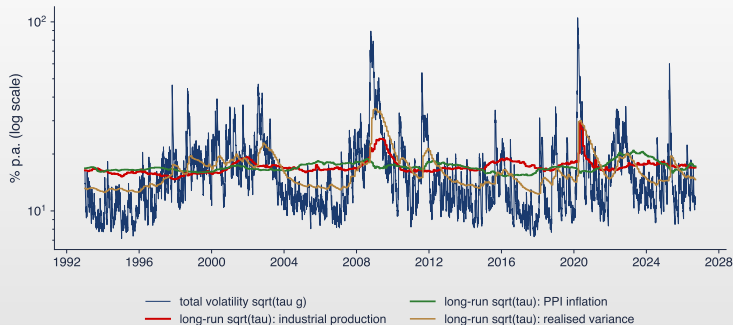
$$\text{VR} = \frac{\text{Var}(\ln \tau_t)}{\text{Var}(\ln(\tau_t g_{i,t}))} \in [0, 1]$$

- ▶ VR close to 0: the long-run component is almost flat; close to 1: it carries most of the variation
 - ▶ a macro variable can be significant ($\hat{\theta} \neq 0$) and still explain little of daily volatility (small VR)
- Inference on w is nonstandard when $\theta = 0$ (w is then not identified) and at the boundary $w = 1$
- Extensions: two-sided weights, several regressors, a second (daily) component [Conrad and Kleen \(2020\)](#); forecasts beyond one month use only the long-run part

Case study: Engle, Ghysels and Sohn (2013) on our data

- ▣ **Question** of [Engle, Ghysels and Sohn \(2013\)](#): does the stock market's long-run volatility move with macroeconomic fundamentals (output growth, inflation)?
 - ▶ their sample of US stock returns is much longer than ours; their benchmark long-run component is a fixed-window realised variance
- ▣ **Our replication**: S&P 500 daily returns, common sample from 04.01.1993, $T = 8482$; $K = 36$ monthly lags; restricted beta weights
 - ▶ three long-run drivers: monthly realised variance (sum of squared daily returns), industrial production growth (FRED INDPRO), PPI inflation (FRED WPSFD49207)
 - ▶ all models against GARCH(1,1) on the same days; QML with sandwich standard errors
- ▣ Extension for a project: Romanian industrial production and the BET (Seminar 8, B5); real-time vintages of the macro data

GARCH-MIDAS for the S&P 500



□ Annualised $\sqrt{252\tau_t g_{i,t}}$ of the industrial-production model and the long-run components $\sqrt{252\tau_t}$ of the three models

GARCH-MIDAS estimates

Long-run driver	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\theta}$ (s.e.)	$\hat{w}$	VR (%)	BIC
realised variance (levels)	0.119	0.841	0.0225 (0.0058)	5.939	25.1	22710.0
industrial production growth	0.110	0.870	-0.551 (0.353)	3.692	4.5	22732.5
PPI inflation	0.108	0.873	0.753 (1.260)	1.000	2.2	22733.3
GARCH(1,1)						22721.9

- Same days for all models; $\text{BIC} = -2 \ln L + k \ln T$ (lower is better); s.e.: Bollerslev–Wooldridge

Interpreting GARCH-MIDAS

- ▣ Realised variance as the slow driver: $\hat{\theta} = 0.0225$ ($t = 3.86$), $VR = 25.1\%$, BIC below GARCH(1,1): a long-run component exists
- ▣ Industrial production: $\hat{\theta} = -0.551$ ($t = -1.56$), the countercyclical sign of [Engle, Ghysels and Sohn \(2013\)](#), but VR only 4.5% and a higher BIC than GARCH(1,1)
- ▣ PPI inflation: $\hat{w}$ at the boundary 1 (flat weights) and $\hat{\theta}$ imprecise: no evidence in 1993–2026
- ▣ Replication verdict: the qualitative sign survives, the strength does not; macro effects need long samples with deep recessions, which is why the original study uses a much longer history

Recap: Long-run components

- ▣ Volatility has at least two speeds; a single GARCH persistence averages them
- ▣ GARCH-MIDAS lets a monthly variable drive the long-run level; identification comes from the mixed frequencies
- ▣ Report VR and BIC next to $\hat{\theta}$: significance is not importance

Prices as Itô semimartingales (1/2): the model

- Within day t , the log price moves by a drift, a Brownian part scaled by the spot volatility, and jumps

$$dX_s = \mu_s ds + \sigma_s dW_s + dJ_s, \quad s \in [0, 1]$$

- ▶ X_s : log price at intraday time s (the day is rescaled to $[0, 1]$); μ_s : drift; W_s : standard Brownian motion
 - ▶ σ_s : stochastic spot (instantaneous) volatility; J_s : a jump process with finitely many jumps per day, ΔJ_s : the jump at time s
- Quadratic variation of the day: the diffusive part plus the squared jumps

$$QV_t = IV_t + \sum_{s \leq 1} (\Delta J_s)^2, \quad IV_t = \int_0^1 \sigma_s^2 ds$$

- ▶ IV_t : integrated variance, the average of the spot variance over the day
- ▶ without jumps and with σ independent of W : $r_t | IV_t \sim N(\int_0^1 \mu_s ds, IV_t)$, so IV is the variance that matters for the daily return r_t

Prices as Itô semimartingales (2/2): realised variance

- Realised variance: the sum of the n squared intraday returns of the day

$$\text{RV}_t = \sum_{i=1}^n r_{t,i}^2, \quad r_{t,i} = X_{i/n} - X_{(i-1)/n}$$

- ▶ n : number of intraday intervals (78 five-minute returns in a 6.5-hour session); $r_{t,i}$: log return over the i -th interval
- ▶ $\text{RV}_t \rightarrow \text{QV}_t$ in probability as $n \rightarrow \infty$: RV measures the total (diffusive plus jump) variation
- Realised variance: the sum of the n squared intraday returns of the day
- ▶ The drift does not matter in the limit: it is $O(1/n)$ per interval, while the Brownian part is $O(n^{-1/2})$
- ▶ volatility becomes observable *ex post*, without a model Andersen, Bollerslev, Diebold and Labys (2003), Barndorff-Nielsen and Shephard (2002)

The central limit theorem for realised variance (1/2)

- Barndorff-Nielsen and Shephard (2002): without jumps, the error of RV shrinks at rate $\sqrt{n}$ and is mixed normal

$$\sqrt{n}(\text{RV}_t - \text{IV}_t) \rightarrow MN(0, 2\text{IQ}_t), \quad \text{IQ}_t = \int_0^1 \sigma_s^4 ds$$

- ▶ IQ_t : integrated quarticity, the daily average of σ_s^4 ; it sets the size of the error
 - ▶ MN (mixed normal): normal given the path of σ , with a random variance 2IQ_t
 - ▶ the convergence is *stable* in law: one may divide by a random, consistently estimated $\sqrt{\text{IQ}_t}$ and keep the limit
- Rate $\sqrt{n}$: with $n = 78$ and volatility constant within the day, the relative error is about $\sqrt{2/78} \approx 16\%$
- ▶ the measurement error $\text{RV}_t - \text{IV}_t$ is heteroskedastic (it scales with IQ_t): HARQ, below, is built on this fact

The central limit theorem for realised variance (2/2): feasible intervals

- ▣ Feasible version: estimate IQ by the realised quarticity RQ, then studentise

$$\widehat{\text{IQ}}_t = \text{RQ}_t = \frac{n}{3} \sum_{i=1}^n r_{t,i}^4, \quad \frac{\text{RV}_t - \text{IV}_t}{\sqrt{\frac{2}{3} \sum_i r_{t,i}^4}} \rightarrow N(0, 1)$$

- ▶ the factor $\frac{n}{3}$ makes RQ unbiased under constant volatility, because $\mathbb{E}Z^4 = 3$ for $Z \sim N(0, 1)$
- ▶ 95% interval for IV_t : $\text{RV}_t \pm 1.96 \sqrt{\frac{2}{3} \sum_i r_{t,i}^4}$; it may contain negative values

- ▣ Log version (delta method): an interval for $\ln \text{IV}_t$

$$\ln \text{RV}_t \pm 1.96 \frac{\sqrt{\frac{2}{3} \sum_i r_{t,i}^4}}{\text{RV}_t}$$

- ▶ exponentiated, it never contains negative values and has better coverage in finite samples

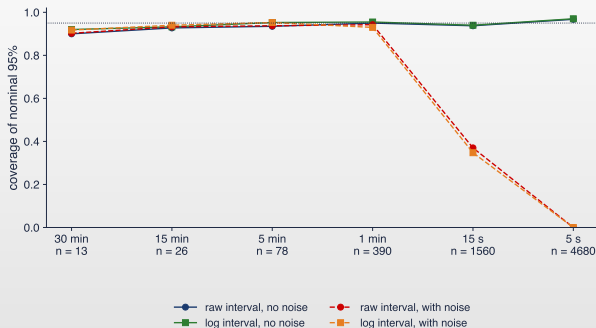
A simulated market with a known truth

- One-second grid, $n = 23\,400$ seconds a day; the log spot variance is the sum of two Gaussian AR(1) factors

$$\ln \sigma_s^2 = c + f_s^{(1)} + f_s^{(2)}$$

- ▶ c : constant fixing the mean; $f^{(1)}$: slow factor (half-life 60 days, stationary s.d. 0.75); $f^{(2)}$: fast factor (half-life 2 days, s.d. 0.45)
- ▶ mean IV = 1 (%² a day), the mean of the S&P 500 realised kernel in 2000–2022; leverage: correlation -0.6 between price shocks and shocks to the fast factor
- Jumps: Poisson with 0.08 jumps a day, sizes $N(0, 0.8^2)$ (%): jumps make about 5.6% of quadratic variation
- Noise: we observe $Y_s = X_s + u_s$, with u_s i.i.d. $N(0, \omega^2)$ and $\omega = 0.004\%$
 - ▶ Y_s : observed log price; X_s : efficient log price; u_s : microstructure noise with standard deviation ω ; at one second, $2n\omega^2 = 0.75$ (next slides)
- Each estimator below is compared with the true IV or QV of the same simulated day

Coverage of the feasible CLT



- 400 simulated days without jumps; 95% intervals for IV_t (raw and log) from returns sampled every 30 minutes to every 5 seconds, without and with noise

Interpreting the CLT coverage

- Without noise, coverage approaches 95% as n grows: raw interval 90.0% at $n = 13$ and 95.0% at $n = 390$; the log interval is closer at small n (92.0%)
- With noise, coverage collapses once sampling is too fine: 37.0% at 15 seconds and 0.0% at 5 seconds; at 1 minute it is still 94.2%
- The interval shrinks like $n^{-1/2}$ while the noise bias grows like n : the CLT is a statement about the efficient price, not about the observed one
- Practical rule: 5-minute returns for liquid assets unless a noise-robust estimator is used

Where the realised measures come from



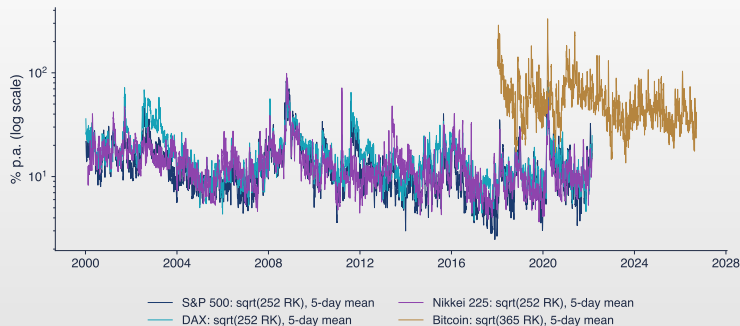
Ole E. Barndorff-Nielsen, 2007

Photo: Thomas Steiner (2007); CC BY-SA 2.5;

[Wikimedia Commons](#)

- [Heber, Lunde, Shephard and Sheppard \(2009\)](#), version 0.3: daily RV, BV, realised kernels and open-to-close returns for 31 indices, 2000 to 25 February 2022, built from cleaned tick data
- its terms: free use if the library and its version are cited; the archived copy keeps the 2022 file available after the shutdown
- Bitcoin and Ether trade 24/7 and Binance publishes all one-minute and one-second prices: we compute RV, BV, RQ, tripower quarticity and a realised kernel for each UTC day, 2018–2026
- No free source covers equity intraday data after 2022: a project can rebuild them from a licensed feed

Realised volatility across markets



- Annualised realised kernel, 5-day means: S&P 500, DAX and Nikkei 225 (Oxford-Man, 2000–2022); Bitcoin (Binance one-minute prices, 2018–2026, 365 days a year)

Interpreting the realised volatilities

- Average annualised volatility from the kernel: S&P 500 16.2%, DAX 19.5%, Nikkei 16.6%, Bitcoin 68.4%
- The equity series move together (2002, 2008, 2011, 2020): a common long-run factor, the multivariate theme of the last section
- Open-to-close measures miss the overnight return: for daily risk one adds the squared overnight return or scales the measure
- Bitcoin has no overnight gap, but its daily volatility is three to four times higher and falls after 2023

Microstructure noise (1/2): the bias of RV

- The observed log price is the efficient price plus a noise term

$$Y_{i/n} = X_{i/n} + u_i$$

- ▶ u_i : i.i.d. noise with variance ω^2 , independent of X ; sources: bid-ask bounce, price discreteness, stale quotes
- RV computed from the observed prices at n intervals, $RV_t^{(n)}$, is biased upwards, and the bias grows with n

$$\mathbb{E}(RV_t^{(n)}|X) = IV_t + 2n\omega^2, \quad \text{Var}(RV_t^{(n)}|X) \approx 4n\mathbb{E}u^4$$

- ▶ each observed return contains $u_i - u_{i-1}$, with variance $2\omega^2$; summed over n returns this gives $2n\omega^2$
- ▶ RV diverges as $n \rightarrow \infty$: sampling as often as possible is not optimal
- ▶ derivation of the bias and of the variance: [▶ Appendix](#)

Microstructure noise (2/2): diagnostics

- Two consequences of i.i.d. noise
 - ▶ $\hat{\omega}^2 = \text{RV}_t^{(n)} / (2n)$ at the highest frequency estimates the noise variance, since $2n\omega^2$ dominates IV there
 - ▶ returns of the observed price have first-order autocovariance $-\omega^2$: negative autocorrelation is the signature of i.i.d. noise
- Signature plot: average $\text{RV}^{(n)}$ against the sampling interval
 - ▶ a flat region shows the intervals where noise no longer matters
- Real noise is not i.i.d.: it is autocorrelated and correlated with the efficient price, especially in quote data
Hansen and Lunde (2006b)

Optimal sparse sampling

- Mean square error (MSE) of $RV^{(n)}$ as an estimator of IV: sampling variance falls, noise bias rises with n

$$MSE(n) \approx \underbrace{\frac{2IQ}{n}}_{\text{variance}} + \underbrace{(2n\omega^2)^2}_{\text{bias}^2} + 4n\mathbb{E}u^4 + \dots$$

- Minimising the first two terms gives the optimal number of intervals [Bandi and Russell \(2008\)](#)

$$n^* \approx \left(\frac{IQ}{4\omega^4} \right)^{1/3}$$

- ▶ the optimal grid is coarser when noise (ω^2) is large relative to volatility (IQ)
- ▶ example (Seminar 8, A3): $IV = IQ = 1$, $\omega^2 = 1.6 \times 10^{-5}$ gives $n^* \approx 990$, a return every 24 seconds
- The best sparse RV converges only at rate $n^{1/6}$ and discards most data: this motivates estimators that use all observations
- Subsampling: average the RV of the K offset grids (starting at second $0, 1, \dots, K-1$); same bias, smaller variance

Two scales and realised kernels (1/2)

- Two-scales RV [Zhang, Mykland and Aït-Sahalia \(2005\)](#): the fast scale estimates the noise bias of the slow one and removes it

$$\text{TSRV} = \overline{\text{RV}}^{(K)} - \frac{\bar{n}}{n} \text{RV}^{(n)}, \quad \bar{n} = \frac{n - K + 1}{K}$$

- ▶ $\overline{\text{RV}}^{(K)}$: subsampled RV on K offset sparse grids (slow scale), each with about $\bar{n}$ returns; $\text{RV}^{(n)}$: RV on all n returns (fast scale)
- ▶ the bias of the slow scale is $2\bar{n}\omega^2 = \frac{\bar{n}}{n} \cdot 2n\omega^2$: the correction subtracts it; rate $n^{1/6}$
- Pre-averaging [Jacod et al. \(2009\)](#): average returns over blocks before squaring
 - ▶ rate $n^{1/4}$, the optimal one in the presence of noise; in practice similar to the realised kernel

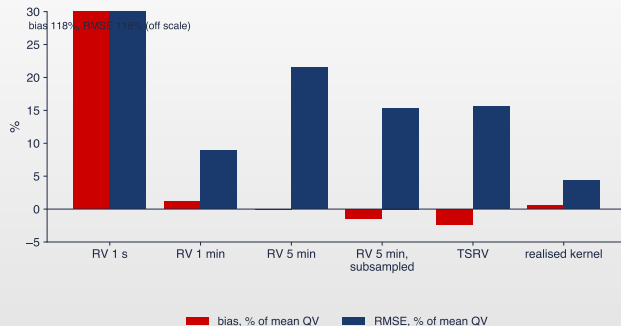
Two scales and realised kernels (2/2): the realised kernel

- Realised kernel [Barndorff-Nielsen, Hansen, Lunde and Shephard \(2008\)](#): RV plus weighted realised autocovariances

$$RK_t = \gamma_0 + \sum_{h=1}^H k\left(\frac{h}{H+1}\right)(\gamma_h + \gamma_{-h}), \quad \gamma_h = \sum_i r_{t,i} r_{t,i-h}$$

- γ_h : realised autocovariance at lag h ($\gamma_0 = \text{RV}$); H : bandwidth, the number of lags used; $k(\cdot)$: weight function with $k(0) = 1$, $k(1) = 0$
 - Parzen weights: $k(x) = 1 - 6x^2 + 6x^3$ for $x \leq \frac{1}{2}$ and $2(1-x)^3$ for $\frac{1}{2} < x \leq 1$
 - the negative γ_1 created by noise cancels the bias $2n\omega^2$: the same idea as HAC variance estimation ([Chapter 0](#))
- Properties
 - nonnegative by construction; rate $n^{1/5}$ with $H = c^* \xi^{4/5} n^{3/5}$, where $\xi^2 = \omega^2 / \sqrt{IQ}$ is the noise-to-signal ratio and $c^* = 3.51$ for Parzen [Barndorff-Nielsen et al. \(2009\)](#)
 - it also corrects serially dependent noise of short range, of either sign

Estimators against the true quadratic variation

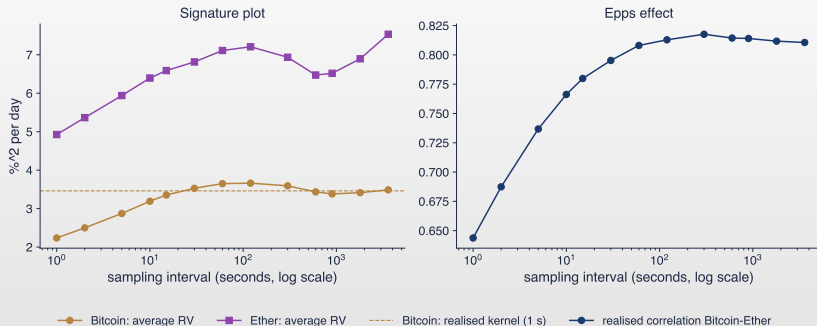


- 300 simulated days with jumps and noise; bias and root mean square error relative to the mean QV; TSRV with $K = 300$ seconds; RK with the BNHLS bandwidth (median $H = 31$)

Interpreting the estimator comparison

- ▣ RV at one second is dominated by noise: bias 117.8% of QV; at one minute the bias is 1.2% and the RMSE 8.9%
- ▣ Five-minute RV is unbiased but noisy (RMSE 21.6%); subsampling and TSRV reduce it to 15.4% and 15.6%
- ▣ The realised kernel uses all 23 400 returns: bias 0.5%, RMSE 4.4%, the best of the six
- ▣ All estimators here target QV, jumps included; separating the jump part needs the next section

Signature plot and Epps effect: Bitcoin and Ether



- Binance one-second prices, August 2026 (31 days); average daily RV by sampling interval (subsampled); realised correlation of the two coins by interval

Interpreting the signature plot

- Bitcoin: RV is 2.24 at one second, 3.65 at one minute, 3.59 at five minutes: the signature *rises*, the opposite of the i.i.d.-noise prediction
- Reason: 52% of one-second returns are zero (no trade or the same price) and prices adjust gradually, so one-second returns are positively autocorrelated
- The realised kernel on one-second data (3.46, median $H = 22$) adds the positive autocovariances back and lands on the flat region of 1–30 minutes
- Epps effect [Epps \(1979\)](#): realised correlation 0.64 at one second against 0.82 at five minutes; asynchronous trading biases covariances towards zero at high frequency

Recap: Realised measures

- ▣ RV estimates QV with error $\sqrt{2IQ/n}$; the log interval behaves better; stable convergence makes studentization valid
- ▣ Noise biases RV at high frequency; its sign and size are empirical: look at the signature plot first
- ▣ Realised kernels use all the data and correct short-range dependent noise; 5-minute RV remains a robust benchmark

Bipower variation (1/2)

- Barndorff-Nielsen and Shephard (2004): sum of products of adjacent absolute returns, rescaled

$$BV_t = \mu_1^{-2} \frac{n}{n-1} \sum_{i=2}^n |r_{t,i}| |r_{t,i-1}|, \quad \mu_1 = \mathbb{E}|Z| = \sqrt{2/\pi}$$

- ▶ $Z \sim N(0, 1)$; μ_1^{-2} makes BV_t consistent for IV_t ; $\frac{n}{n-1}$ corrects for the $n-1$ products
- ▶ $BV_t \rightarrow IV_t$ even with (finite-activity) jumps: a jump enters only two products, each multiplied by a return of order $n^{-1/2}$
- Hence $RV_t - BV_t \rightarrow \sum_{s \leq 1} (\Delta J_s)^2$: a nonparametric estimate of the jump variation of the day

Bipower variation (2/2): related estimators

- ▣ Jump-robust quarticity: the tripower quarticity

$$\text{TQ}_t = n \mu_{4/3}^{-3} \frac{n}{n-2} \sum_{i=3}^n \prod_{j=0}^2 |r_{t,i-j}|^{4/3}, \quad \mu_{4/3} = \mathbb{E}|Z|^{4/3}$$

- ▶ products of three adjacent absolute returns, each to the power $4/3$; $\text{TQ}_t \rightarrow \text{IQ}_t$ even with jumps (RQ does not)
- ▣ Nearest-neighbour truncation (MedRV, MinRV) Andersen, Dobrev and Schaumburg (2012): smaller finite-sample bias from a jump and from zero returns
- ▣ Threshold (truncated) RV: drop the returns above $c n^{-\varpi}$
 - ▶ $c > 0$: a constant (a multiple of the local volatility); $\varpi \in (0, \frac{1}{2})$: the threshold shrinks more slowly than a diffusive return, so only jumps are removed

Testing for jumps (1/2): the ratio statistic

- Barndorff-Nielsen and Shephard (2006): without jumps, the difference $RV - BV$ is also mixed normal

$$\sqrt{n}(RV_t - BV_t) \rightarrow MN(0, \vartheta IQ_t), \quad \vartheta = \mu_1^{-4} + 2\mu_1^{-2} - 5 = \frac{\pi^2}{4} + \pi - 5 \approx 0.609$$

- ▶ ϑ : a constant that measures how much less efficient BV is than RV
- Ratio statistic Huang and Tauchen (2005): the relative jump contribution, studentised

$$z_t = \frac{(RV_t - BV_t)/RV_t}{\sqrt{\frac{\vartheta}{n} \max(1, TQ_t/BV_t^2)}} \rightarrow N(0, 1)$$

- ▶ one-sided test: reject “no jump on day t ” for large z_t
- ▶ the ratio form and the max adjustment give the best size in their simulations; $TQ_t/BV_t^2 \geq 1$ by Jensen when volatility varies within the day

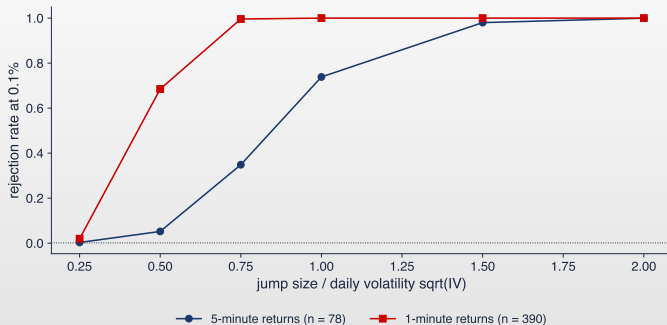
Testing for jumps (2/2): in practice

- A daily test answers “was there a jump today?”; locating it within the day needs a test on each return standardised by local BV [Lee and Mykland \(2008\)](#)
- Multiple testing: at level α over T days one expects αT false jump days
 - ▶ use a small α (0.1%) or a family-wise (FWER) or false-discovery-rate (FDR) correction ([Chapter 1](#))
- Separating the continuous and jump parts of RV [Andersen, Bollerslev and Diebold \(2007\)](#)

$$J_t = \mathbb{I}(z_t > z_{1-\alpha})(RV_t - BV_t), \quad C_t = RV_t - J_t$$

- ▶ $\mathbb{I}(\cdot)$: indicator, 1 if the test rejects and 0 otherwise; $z_{1-\alpha}$: the $1 - \alpha$ quantile of $N(0, 1)$
- ▶ J_t : jump part, nonzero only on significant days; C_t : continuous part

Size and power of the ratio test

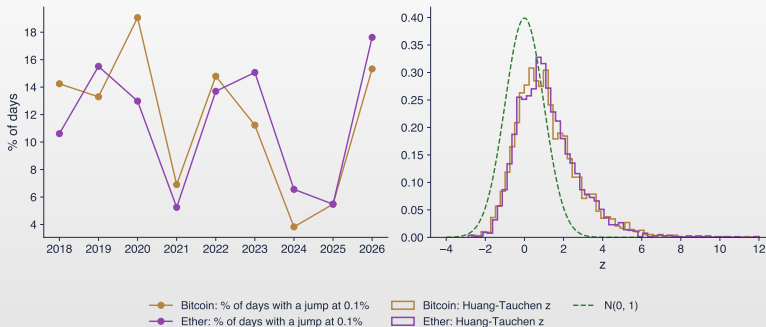


- 2000 simulated days (two-factor SV, no jumps), then one jump of size $c\sqrt{IV_t}$ at a random time; one-sided test at 0.1% with 5-minute and 1-minute returns

Interpreting the size and power

- Size without jumps: 0.0% (5 minutes) and 0.2% (1 minute) against the nominal 0.1%; with i.i.d. noise of the calibrated size still 0.1% at 1 minute
- Power for a jump of half a daily standard deviation: 5% with 5-minute returns, 68% with 1-minute returns; for one standard deviation 74% and 100%
- The power depends on $c\sqrt{n}$: small jumps are absorbed into the diffusion at coarse sampling (Seminar 8, A6 gives the detection threshold)
- The trade-off with noise is the same as for RV: finer sampling raises power until noise and zero returns distort the statistic

Jump days of Bitcoin and Ether



- Huang–Tauchen test with 5-minute returns ($n = 288$ a UTC day), 2018–2026, $T = 3\,165$ days; left: share of jump days at 0.1% by year; right: distribution of z_t against $N(0, 1)$

Interpreting the crypto jump tests

- ▣ Bitcoin: 362 jump days (11.4%) where about 3.2 false rejections are expected; Ether: 355 days (11.2%)
- ▣ Yet the jumps are small: they make 3.0% (Bitcoin) and 2.3% (Ether) of total variation; the rest is continuous
- ▣ The whole distribution of z_t is shifted (mean 1.16, not 0): besides jumps, 24-hour intraday seasonality, zero returns and flash moves violate the assumptions of the null
- ▣ Before calling a day a jump day: standardise returns by an intraday volatility pattern, check zero returns, and control the number of tests

Recap: Jumps

- BV estimates the continuous part; $RV - BV$ the jump part; TQ makes the test robust to jumps in the quarticity
- The ratio test has good size in simulations; its power grows with $c\sqrt{n}$
- In real data the null is fragile: many rejections, little jump variation; treat jump counts as model-dependent

The HAR model of Corsi (1/2): the regression

- Corsi (2009): tomorrow's RV is a linear function of the daily, weekly and monthly averages of past RV

$$RV_{t+1} = \beta_0 + \beta_d RV_t + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + u_{t+1}$$

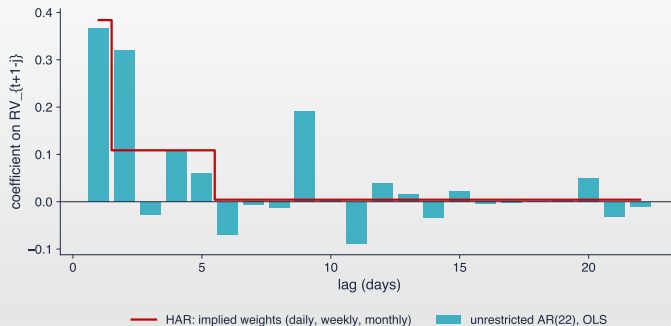
$$RV_t^{(w)} = \frac{1}{5} \sum_{j=0}^4 RV_{t-j}, \quad RV_t^{(m)} = \frac{1}{22} \sum_{j=0}^{21} RV_{t-j}$$

- ▶ $RV_t^{(w)}, RV_t^{(m)}$: averages over the last 5 trading days (a week) and the last 22 (a month)
- ▶ $\beta_d, \beta_w, \beta_m \geq 0$: weights of the three horizons; β_0 : intercept; u_{t+1} : forecast error with mean zero
- ▶ $\beta_d + \beta_w + \beta_m$: the persistence; the closer to 1, the slower volatility returns to its mean
- Heterogeneous market hypothesis: daily, weekly and monthly traders react to volatility at their own horizon
 - ▶ the cascade produces slowly decaying autocorrelations

The HAR model of Corsi (2/2): estimation and variants

- ▣ Statistically, HAR is an AR(22) with 19 linear restrictions (step-shaped coefficients)
 - ▶ short memory, but it mimics long memory over a month
- ▣ OLS is consistent; the errors are heteroskedastic and, for multi-day targets, overlapping
 - ▶ use Newey–West [Newey and West \(1987\)](#) (HAC) standard errors
- ▣ Variants
 - ▶ log-HAR: the same regression on $\ln RV$, with near-Gaussian errors; the forecast of RV is $\exp(\hat{\mu} + \frac{1}{2}\hat{\sigma}_u^2)$, with $\hat{\mu}$ the fitted log value and $\hat{\sigma}_u^2$ the residual variance
 - ▶ HAR on $\sqrt{RV}$; h -day targets (the average RV over the next h days)
- ▣ Extensions: continuous and jump parts, HAR-CJ [Andersen, Bollerslev and Diebold \(2007\)](#); positive and negative semivariances [Patton and Sheppard \(2015\)](#); implied volatility; leverage terms

HAR as a restricted AR(22)



- S&P 500 (Oxford-Man, 5-minute RV, 2000–2022, $T = 5\,530$): coefficients of an unrestricted AR(22) by OLS and the step weights implied by the HAR estimates

Interpreting the HAR estimates

- ▣ HAR: $\hat{\beta}_d = 0.275$ (0.104), $\hat{\beta}_w = 0.522$ (0.140), $\hat{\beta}_m = 0.094$ (0.078), Newey–West s.e.; $R^2 = 0.560$; persistence $\sum \hat{\beta} = 0.891$
- ▣ The AR(22) with 23 coefficients reaches $R^2 = 0.596$ in sample, with erratic coefficients; HAR captures the shape with 4
- ▣ Log-HAR: $\hat{\beta}_d = 0.393$, $\hat{\beta}_w = 0.399$, $\hat{\beta}_m = 0.153$, $R^2 = 0.719$ on $\ln RV$; in logs the monthly component matters more
- ▣ In levels a few crisis days dominate OLS: large standard errors on β_d and β_m ; weighted least squares or logs give more stable estimates

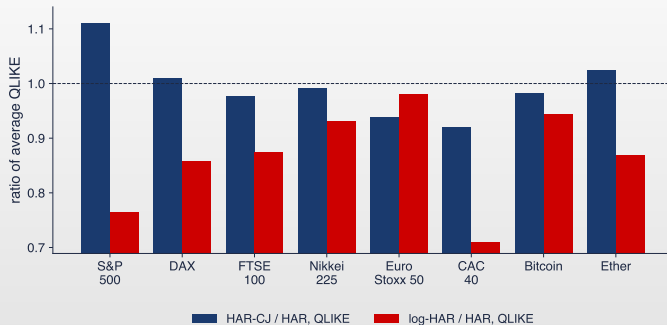
Forecasting design

- ▣ One day ahead, rolling window of 1 000 days, re-estimated every day; target RV_{t+1}
- ▣ Losses: $MSE = (RV - f)^2$ and QLIKE, both robust to the noise in RV (see below)

$$QLIKE = \frac{RV}{f} - \ln \frac{RV}{f} - 1 \geq 0$$

- ▶ f : the variance forecast; QLIKE is 0 when $f = RV$ and penalises under-prediction more than over-prediction
- ▣ Insanity filter [Bollerslev, Patton and Quaadvlieg \(2016\)](#): a forecast outside the range of the in-sample RV is replaced by the in-sample mean
 - ▶ linear HAR-type forecasts can be negative after a spike; QLIKE is then undefined
 - ▶ the filter is part of the method and must be reported
- ▣ Diebold–Mariano [Diebold and Mariano \(1995\)](#) t -statistic of the loss differential against HAR
 - ▶ Newey–West variance; negative = better than HAR, below -1.96 significant at 5%; several models: the model confidence set [Hansen, Lunde and Nason \(2011\)](#), Chapter 1
- ▣ Models: HAR, HAR-CJ (continuous part and jump part, from BV), log-HAR; for Bitcoin and Ether also HARQ

HAR extensions out of sample



- Average QLIKE relative to HAR (below 1 = better), six Oxford-Man indices (2004–2022) and two cryptocurrencies (from 01.11.2020); 1 000-day rolling window

Interpreting the out-of-sample HAR comparison

- Log-HAR beats HAR in QLIKE for 6 of 6 indices: S&P 500 0.764 (DM -2.07), CAC 40 0.709 (DM -1.73), FTSE 0.874 (DM -5.08)
- HAR-CJ helps in 4 of 6 indices, but not for the S&P 500 (1.110): separating jumps adds estimation noise when jumps are small
- The ranking depends on the loss: in MSE the gains of log-HAR are smaller (S&P 500: 0.837); MSE weighs the crisis days most
- Six indices from one data vendor and one period: report all of them, not the best one (data snooping, [Chapter 1](#))

HARQ (1/2): the measurement error of RV

- RV is the latent IV plus a measurement error whose variance changes every day

$$RV_t = IV_t + e_t, \quad \text{Var}(e_t|IQ_t) \approx \frac{2 IQ_t}{n}$$

- ▶ e_t : measurement error of day t ; on volatile days (IQ_t large) RV is less precise

- Errors in variables: in an AR(1) for IV with coefficient ϕ , the OLS slope on RV is attenuated (▶ [Appendix](#))

$$\text{plim } \hat{\phi} = \phi\lambda, \quad \lambda = \frac{\text{Var}(IV)}{\text{Var}(IV) + \mathbb{E}\text{Var}(e)} \in (0, 1)$$

- ▶ λ : reliability ratio, the share of the variance of RV that is signal
- ▶ one constant β_d is a compromise: too high on noisy days, too low on precise days

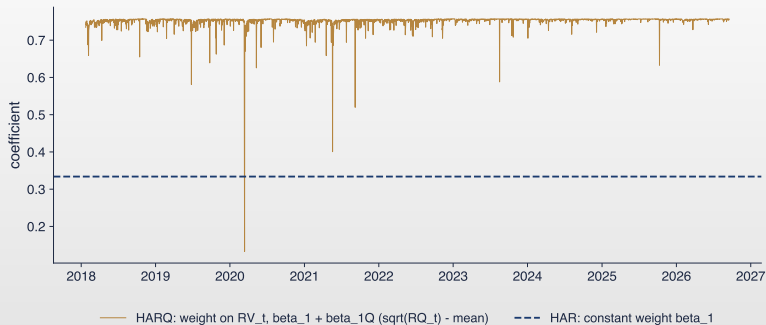
HARQ (2/2): a weight that depends on precision

- Bollerslev, Patton and Quaadvlieg (2016): the weight of yesterday's RV moves with its estimated precision

$$RV_{t+1} = \beta_0 + (\beta_d + \beta_{dQ} \sqrt{RQ_t}) RV_t + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + u_{t+1}$$

- ▶ RQ_t : realised quarticity, the estimate of IQ_t ; $\sqrt{RQ_t}$ is proportional to the standard deviation of e_t
- ▶ $\beta_{dQ} < 0$ expected: the less precise yesterday's RV, the lower its weight
- ▶ $\sqrt{RQ_t}$ is demeaned, so β_d is the weight on a day of average precision
- One extra parameter, still OLS
- The original paper: S&P 500 futures and 27 Dow Jones stocks, gains over HAR in and out of sample
 - ▶ our replication extends it to Bitcoin and Ether, 2018–2026, where RQ is available from our one-minute data

HARQ for Bitcoin: a weight that moves with precision



- Full-sample HARQ on Bitcoin 5-minute RV and RQ: daily weight $\hat{\beta}_d + \hat{\beta}_{dQ}(\sqrt{RQ_t} - \sqrt{RQ})$ against the constant HAR weight

Interpreting HARQ

- ▣ $\hat{\beta}_{dQ} = -0.00021$ ($t = -2.7$, Newey–West): negative, as predicted; the daily weight at average precision is 0.752, against 0.334 in HAR
- ▣ On most days the weight is near 0.76; on the noisiest days it falls (1% quantile 0.71, minimum 0.13): yesterday's RV is trusted only when it is precise
- ▣ Out of sample, QLIKE relative to HAR: Bitcoin 0.946 (DM -2.44), Ether 0.921 (DM -4.50): the gain of the original study carries over to crypto
- ▣ In MSE the picture differs (Bitcoin 1.163): MSE is dominated by a few extreme days, where HARQ shrinks the most; the AI mini-case checks how fragile the gain is

Recap: HAR and HARQ

- ▣ HAR is a parsimonious restricted AR(22) estimated by OLS; logs stabilise it
- ▣ Out-of-sample gains are loss- and market-specific: report all series, the filter and DM tests
- ▣ HARQ lets the measurement error of RV set the weight of yesterday: one parameter, a robust gain in QLIKE

Realized GARCH (1/2): the three equations

- Hansen, Huang and Shek (2012), log-linear form: a return equation, a GARCH equation driven by the realised measure, and a measurement equation

$$r_t = \sqrt{h_t} z_t, \quad \ln h_t = \omega + \beta \ln h_{t-1} + \gamma \ln x_{t-1}$$

$$\ln x_t = \xi + \varphi \ln h_t + \tau(z_t) + u_t$$

- ▶ h_t : conditional variance of the return r_t ; z_t : i.i.d. standardised shock; x_t : realised measure of day t (here the realised kernel RK)
- ▶ β : persistence of $\ln h_t$; γ : weight of yesterday's realised measure (the news variable that replaces ε_{t-1}^2)
- ▶ u_t : measurement error, i.i.d. $N(0, \sigma_u^2)$, independent of z_t
- Measurement equation: x_t is a noisy, possibly biased signal of h_t
 - ▶ $\varphi = 1$: x_t proportional to h_t ; ξ absorbs the scale (open-to-close measure against close-to-close variance)
 - ▶ leverage function $\tau(z) = \tau_1 z + \tau_2(z^2 - 1)$: with $\tau_1 < 0$, negative returns raise the next realised measure (news impact, TSA, Chapter 5)

Realized GARCH (2/2): reduced form and estimation

- ▣ Substituting the measurement equation into the GARCH equation: $\ln h_t$ is an AR(1)

$$\ln h_t = (\omega + \gamma\xi) + \pi \ln h_{t-1} + \gamma(\tau(z_{t-1}) + u_{t-1}), \quad \pi = \beta + \varphi\gamma$$

- ▶ π : persistence of the log variance; multi-step forecasts follow, because x_t has its own equation
- ▣ Joint QML: the log-likelihood sums a return part and a measurement part

$$\ell = -\frac{1}{2} \sum_t \left[\underbrace{\ln h_t + z_t^2}_{r_t} + \underbrace{\ln \sigma_u^2 + u_t^2 / \sigma_u^2}_{x_t} \right]$$

- ▶ comparison with GARCH only through the partial likelihood of r_t (the first part), since GARCH does not model x_t

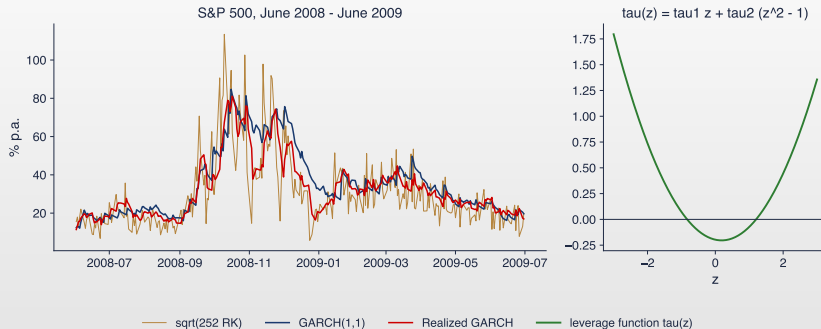
HEAVY and the family of realised-measure models

- HEAVY Shephard and Sheppard (2010): two GARCH-type equations, one for the return variance and one for the mean of the realised measure

$$\begin{aligned}h_t &= \text{Var}(r_t | \mathcal{F}_{t-1}) = \omega + \alpha \text{RM}_{t-1} + \beta h_{t-1} \\ \mu_t &= \mathbb{E}(\text{RM}_t | \mathcal{F}_{t-1}) = \omega_R + \alpha_R \text{RM}_{t-1} + \beta_R \mu_{t-1}\end{aligned}$$

- ▶ RM_t : realised measure of day t ; (ω, α, β) and $(\omega_R, \alpha_R, \beta_R)$: the parameters of the two equations
- ▶ each equation is estimated separately by QML; momentum: after a shock, h_t keeps rising while μ_t catches up
- ▶ when the realised measure is in the variance equation, the squared return usually adds nothing (ε_{t-1}^2 gets a zero coefficient)
- Realized EGARCH Hansen and Huang (2016): several realised measures and a richer leverage; score-driven versions exist for fat tails
- Compared with HAR: returns and realised measures are modelled jointly, so densities and VaR/ES forecasts are available (Chapter 9)

Case study: Realized GARCH for the S&P 500



- Open-to-close returns and the Parzen realised kernel (Oxford-Man), 2000–2022, $T = 5552$, as in Hansen, Huang and Shek (2012); left: the 2008 crisis; right: the estimated leverage function

Realized GARCH estimates

ω	β	γ	ξ	φ	τ_1	τ_2	σ_u
0.119 (0.010)	0.678 (0.018)	0.287 (0.016)	-0.456 (0.021)	1.012 (0.026)	-0.073 (0.008)	0.197 (0.010)	0.671 (0.008)

- Bollerslev–Wooldridge s.e. in brackets; persistence $\hat{\pi} = \hat{\beta} + \hat{\varphi}\hat{\gamma} = 0.968$ (GARCH(1,1) on the same returns: 0.989)
- Partial log-likelihood of the returns: Realized GARCH -6899.7, HEAVY -6902.8, GJR -6998.8, GARCH(1,1) -7099.0
- HEAVY variance equation: $\hat{\alpha} = 0.377$ on RK_{t-1} , $\hat{\beta} = 0.691$

Interpreting Realized GARCH

- ▣ $\hat{\phi} = 1.012$: the kernel is proportional to the conditional variance; $\hat{\xi} = -0.456$ with $\hat{\sigma}_u = 0.671$: on average the kernel lies below the conditional variance of the open-to-close return
- ▣ $\hat{\gamma} = 0.287$ is large and $\hat{\beta} = 0.678$ smaller than in GARCH: the realised measure brings information faster than squared returns (left panel, autumn 2008)
- ▣ Leverage: $\hat{\tau}_1 = -0.073$, $\hat{\tau}_2 = 0.197$: an asymmetric parabola, larger for negative z
- ▣ The gain in the partial likelihood over GARCH(1,1) (199.3 points) has the direction of the original paper; HEAVY is almost as good with three parameters

Recap: Realised-measure GARCH

- ▣ A realised measure in the variance equation beats squared returns as the news variable
- ▣ Realized GARCH adds a measurement equation: multi-step forecasts, leverage and a joint likelihood
- ▣ Compare with GARCH only on the partial likelihood of returns

The proxy problem and robust losses (1/2)

- We never see σ_t^2 : a forecast h_t is compared with a proxy $\hat{\sigma}_t^2$ (squared return, RV, RK)
 - ▶ the proxy is conditionally unbiased, $\mathbb{E}(\hat{\sigma}_t^2 | \mathcal{F}_{t-1}) = \sigma_t^2$, but noisy
- A loss L is *robust* [Patton \(2011\)](#) if the ranking of forecasts by $\mathbb{E}L(\hat{\sigma}_t^2, h_t)$ equals the ranking by $\mathbb{E}L(\sigma_t^2, h_t)$, for every unbiased proxy [Hansen and Lunde \(2006a\)](#)
- Necessary and sufficient condition: a Bregman form ([Chapter 1](#), consistent scoring functions for the mean)

$$L(\hat{\sigma}^2, h) = \tilde{C}(h) + B(\hat{\sigma}^2) + C(h)(\hat{\sigma}^2 - h), \quad C'(h) < 0$$

- ▶ C : a decreasing function of the forecast; $\tilde{C}$: its antiderivative, $\tilde{C}' = C$; B : any function of the proxy alone (it does not affect the ranking)

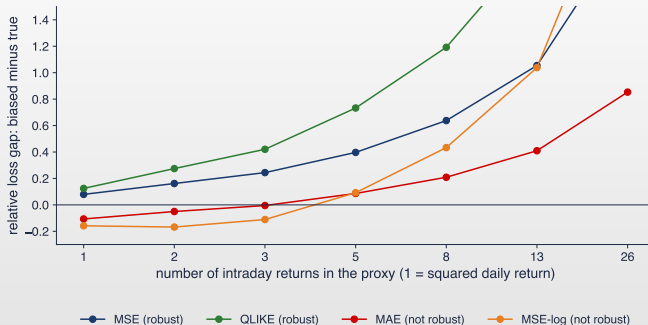
The proxy problem and robust losses (2/2)

- ▣ Robust: MSE and QLIKE, the members of degree 2 and degree 0 of the homogeneous robust family

$$\text{MSE} = (\hat{\sigma}^2 - h)^2, \quad \text{QLIKE} = \frac{\hat{\sigma}^2}{h} - \ln \frac{\hat{\sigma}^2}{h} - 1$$

- ▶ degree: how the loss scales when proxy and forecast are multiplied by the same constant; QLIKE (degree 0) is scale-free and penalises under-prediction more
- ▣ Not robust: MAE = $|\hat{\sigma}^2 - h|$, MSE on logs ($\ln \hat{\sigma}^2 - \ln h$)², MSE on standard deviations
 - ▶ with a noisy proxy they reward forecasts that are biased downwards
- ▣ Example: with $\hat{\sigma}^2 = r^2$, the MSE-log optimum is $h^* = \exp(\mathbb{E} \ln r^2) = 0.28 \sigma^2$
 - ▶ because $r^2 = \sigma^2 \chi_1^2$ for a Gaussian return and $\mathbb{E} \ln \chi_1^2 = -1.27$: MSE-log prefers a forecast far below the truth

Robust and non-robust losses

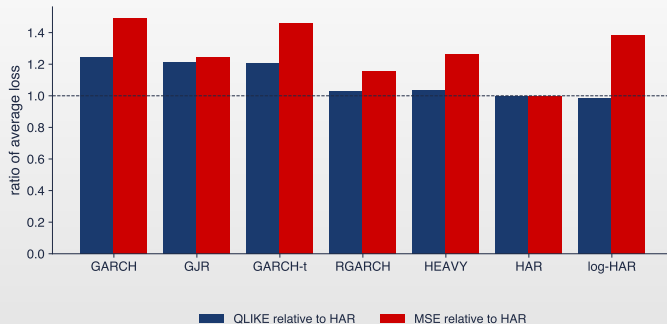


- True variance against a forecast biased down by the factor 0.6; proxy = RV from n intraday returns ($n = 1$: squared daily return); relative gap of expected losses, above zero = the true variance wins

Interpreting the robustness experiment

- ▣ With the squared return as proxy, MAE prefers the biased forecast (gap -10.7%) and so does MSE-log (-15.8%); MSE (7.9%) and QLIKE (12.5%) do not
- ▣ With $n = 5$ intraday returns all four losses rank correctly: a precise proxy makes non-robust losses “almost robust efHLb
- ▣ The danger is largest exactly where realised measures are unavailable: daily data, emerging markets, long histories
- ▣ Rule: evaluate volatility forecasts with QLIKE (and MSE), against the best available proxy

Seven forecasts of S&P 500 volatility



- Open-to-close variance, one day ahead, 2016–2022 (1 537 days); expanding window from 2000, parameters re-estimated every 250 days; proxy: realised kernel

Interpreting the forecast comparison

- ▣ QLIKE: HAR 0.328, log-HAR 0.324, Realized GARCH 0.337, HEAVY 0.339, GARCH- t 0.396, GJR 0.398, GARCH 0.407
- ▣ DM against HAR (QLIKE): GARCH 8.62, GJR 6.27, Realized GARCH 1.42, log-HAR -0.37 : realised-measure models beat return-only models clearly; among them the differences are small
- ▣ In MSE, HAR (1.60) leads and the DM statistics are smaller: the 2020 crash dominates the squared errors
- ▣ One proxy (RK) for all models: the HAR family is fitted to that same measure, which favours it; a fair comparison also tries another proxy (RV, BV)

Recap: Evaluation

- Only losses of the Patton class keep the ranking under a noisy unbiased proxy: MSE and QLIKE
- Realised-measure models beat return-only GARCH one day ahead; DM and the MCS of [Chapter 1](#) quantify by how much
- State the proxy, the loss, the window and the filter: each can change the ranking

From one variance to a covariance matrix (1/2)

- The vector of N return shocks is a matrix square root of the conditional covariance times a standardised vector

$$\varepsilon_t = H_t^{1/2} \eta_t, \quad \eta_t \text{ i.i.d. } (0, I_N)$$

- ▶ $H_t = \text{Var}(\varepsilon_t | \mathcal{F}_{t-1})$: $N \times N$ conditional covariance matrix; I_N : identity matrix
- ▶ H_t must be symmetric positive definite for every t and every parameter value
- VEC: every element of H_t depends on all past squares and cross-products

$$\text{vech}(H_t) = c + A \text{vech}(\varepsilon_{t-1} \varepsilon_{t-1}') + B \text{vech}(H_{t-1})$$

- ▶ vech : stacks the $N(N+1)/2$ distinct elements of a symmetric matrix; c : vector, A, B : square matrices of that size
- ▶ general, but positivity is hard to impose

From one variance to a covariance matrix (2/2): BEKK, CCC and DCC

- BEKK Engle and Kroner (1995): quadratic forms guarantee positive definiteness

$$H_t = CC' + A'\varepsilon_{t-1}\varepsilon'_{t-1}A + B'H_{t-1}B$$

- ▶ C : lower-triangular $N \times N$; A, B : $N \times N$ matrices; diagonal and scalar versions restrict A, B
- ▶ identification: (A, B) and $(-A, -B)$ give the same H_t ; fix the sign of one element

- CCC Bollerslev (1990) and DCC Engle (2002): variances from univariate GARCH, correlations modelled separately

$$H_t = D_t R D_t \quad (\text{CCC}), \quad H_t = D_t R_t D_t \quad (\text{DCC}), \quad D_t = \text{diag}(\sigma_{1t}, \dots, \sigma_{Nt})$$

- ▶ σ_{it} : conditional standard deviation of asset i from its own GARCH; R : constant correlation matrix; R_t : time-varying correlation matrix

The curse of dimensionality in numbers

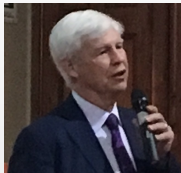
N	VEC	BEKK	diagonal BEKK	scalar BEKK	DCC
2	21	11	7	5	9
5	465	65	25	17	27
10	6 105	255	75	57	77
25	211 575	1 575	375	327	377
50	3 252 525	6 275	1 375	1 277	1 377
100	51 010 050	25 050	5 250	5 052	5 252

- Counts of free parameters of the (1,1) versions; DCC: $3N$ univariate GARCH parameters, 2 correlation parameters and the $N(N-1)/2$ elements of the target
- The DCC target is estimated by moments (correlation targeting), so the numerical search has only 2 parameters at any N ; the price: $N(N-1)/2$ estimated correlations

Asymptotic theory for multivariate GARCH

- Gaussian QML is the standard estimator; consistency and asymptotic normality: VEC and BEKK [Comte and Lieberman \(2003\)](#), [Hafner and Preminger \(2009\)](#); CCC with ARMA means [Ling and McAleer \(2003\)](#)
- Conditions are stronger than in the univariate case (moments of order 6 or 8 for BEKK in early results); the sandwich of the univariate section carries over
- DCC is estimated in two steps [Engle and Sheppard \(2001\)](#): (1) univariate GARCH for each series; (2) QML of the correlation part given the standardised residuals $z_t = D_t^{-1} \varepsilon_t$
 - ▶ second-step standard errors must account for the first step (a GMM-type correction); naive second-step s.e. are too small
- Large N : the full likelihood needs R_t^{-1} and $|R_t|$ at each t ($O(N^3)$) and the score is dominated by noise; composite likelihood over pairs [Pakel, Shephard, Sheppard and Engle \(2021\)](#) avoids both

DCC and its corrected version cDCC (1/2): DCC



Robert F. Engle, 2017

Photo: Econterms (2017); CC BY-SA 4.0;

[Wikimedia Commons](#)

- DCC: a GARCH(1,1)-type recursion for a quasi-correlation matrix Q_t , rescaled into a correlation matrix

$$Q_t = (1 - a - b)S + a z_{t-1} z'_{t-1} + b Q_{t-1}$$

$$R_t = \text{diag}(Q_t)^{-1/2} Q_t \text{diag}(Q_t)^{-1/2}$$

- ▶ $z_t = D_t^{-1} \varepsilon_t$: standardised residuals of the univariate GARCH models; S : their sample correlation matrix (the target)
- ▶ $a \geq 0$: reaction of correlations to yesterday's co-movement; $b \geq 0$: persistence; $a + b < 1$
- ▶ $\text{diag}(Q_t)$: the diagonal matrix of the diagonal elements of Q_t ; the rescaling puts ones on the diagonal of R_t

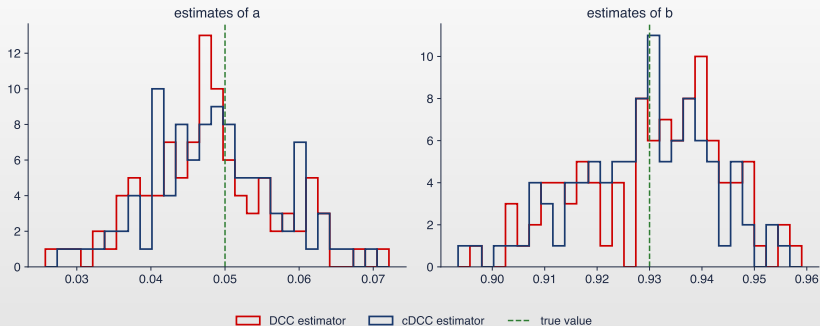
DCC and its corrected version cDCC (2/2): cDCC

- Aielli (2013): the DCC target is estimated inconsistently
 - ▶ $\mathbb{E}(z_t z_t' | \mathcal{F}_{t-1}) = R_t \neq Q_t$, so in general $\mathbb{E}Q_t \neq \mathbb{E}z_t z_t'$
 - ▶ the moment estimator of S is inconsistent, and so is $(\hat{a}, \hat{b})$
- cDCC: rescale the residuals by the diagonal of Q_{t-1} before they enter the recursion

$$z_{t-1}^* = \text{diag}(Q_{t-1})^{1/2} z_{t-1}, \quad Q_t = (1 - a - b)S + a z_{t-1}^* z_{t-1}^{*'} + b Q_{t-1}$$

- ▶ then $\mathbb{E}(z_t^* z_t^{*'} | \mathcal{F}_{t-1}) = Q_t$, and $S = \mathbb{E}z_t^* z_t^{*'}$ is a valid target
- ▶ the cDCC target depends on (a, b) : it is estimated iteratively inside the likelihood

DCC against cDCC: a simulation



- Bivariate cDCC process with $a = 0.05$, $b = 0.93$, target correlation 0.5, unit variances; $T = 2\,000$, 100 replications; both estimators on each sample

Interpreting the DCC simulation

- Mean estimates: DCC $\hat{a} = 0.0478$, $\hat{b} = 0.9307$; cDCC $\hat{a} = 0.0486$, $\hat{b} = 0.9288$ (true 0.05 and 0.93)
- RMSE of $\hat{a}$: 0.0090 (DCC) and 0.0086 (cDCC); of $\hat{b}$: 0.0135 and 0.0129: the inconsistency is real but small at these persistence levels
- Sampling error dominates the bias with two thousand observations; the difference matters more for the target in large systems and for inference on (a, b)
- A proof of inconsistency does not tell you the size of the bias: simulate it in your own design before choosing an estimator

Large covariance matrices: shrinkage and DCC-NL (1/2)

- With N/T not small, the eigenvalues of a sample covariance S are too dispersed
 - ▶ N : number of assets; T : number of observations; the largest eigenvalues are too large, the smallest too small; the inverse S^{-1} amplifies the error
- Global minimum-variance (GMV) portfolio: the weights with the smallest variance that sum to 1

$$w = \frac{\Sigma^{-1} \mathbf{1}}{\mathbf{1}' \Sigma^{-1} \mathbf{1}}$$

- ▶ Σ : covariance matrix of the returns; $\mathbf{1}$: vector of ones
- ▶ it loads on the directions with the smallest, most underestimated eigenvalues
- ▶ it depends only on Σ , so its realised risk measures the quality of the covariance forecast without noise from expected returns

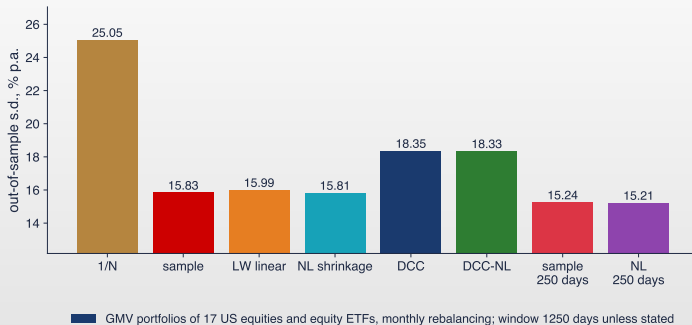
Large covariance matrices: shrinkage and DCC-NL (2/2)

- Linear shrinkage [Ledoit and Wolf \(2004\)](#): a weighted average of S and a scaled identity

$$\hat{\Sigma} = \delta \mu I + (1 - \delta) S$$

- ▶ μ : average eigenvalue of S ; $\delta \in [0, 1]$: shrinkage intensity, estimated from the data (larger when N/T is larger)
- Nonlinear shrinkage [Ledoit and Wolf \(2020\)](#): keep the eigenvectors of S , replace each eigenvalue λ_i by $d(\lambda_i)$
 - ▶ $d(\cdot)$: an analytical kernel formula that pulls small eigenvalues up and large ones down, by different amounts
- DCC-NL [Engle, Ledoit and Wolf \(2019\)](#): in DCC, the target S of the standardised residuals is itself a large sample covariance: replace it by its nonlinear shrinkage
 - ▶ composite likelihood for (a, b) , univariate GARCH for the variances; evaluated by the out-of-sample s.d. of GMV portfolios

Case study: Engle, Ledoit and Wolf (2019) on US equities



- 17 assets: 14 US stocks and three equity ETFs (SPY, QQQ, RSP), which are portfolios of stocks; GMV weights rebalanced every 21 days from 23.05.2017 (112 rebalancings); window 1 250 days, and 250 days for the last two bars; annualised out-of-sample s.d.

Interpreting the GMV comparison (1/2)

- Out-of-sample s.d. (% p.a.): 1/N 25.05, sample 15.83, linear shrinkage 15.99, nonlinear shrinkage 15.81, DCC 18.35, DCC-NL 18.33
- With $N/T = 17/1250$ the sample covariance is already accurate
 - ▶ nonlinear shrinkage changes little (15.81 against 15.83); DCC-NL is no better than DCC
- Here the dynamics hurt: DCC gives 18.35 against 15.83
 - ▶ DCC uses GARCH variances, with forecasts averaged over the 21-day holding period
 - ▶ one-month-ahead variance forecasts of single stocks such as MSTR and TSLA are noisy, and GMV amplifies their errors

Interpreting the GMV comparison (2/2)

- A window of 250 days makes N/T five times larger
 - ▶ s.d. of the sample GMV 15.24, with nonlinear shrinkage 15.21: only 0.2% lower
 - ▶ yet the ETFs make the correlation matrix nearly singular (median condition number 1146)
 - ▶ the shorter window itself lowers the risk: a crude form of time variation
- The original study uses hundreds of stocks, where N/T is large
 - ▶ there the DCC-NL gains are substantial
 - ▶ with our N the lesson is when shrinkage matters, not how much it gains

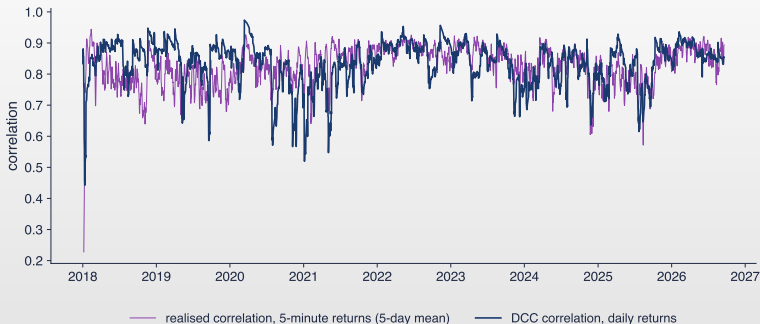
Realised covariance

- The multivariate RV: the sum of outer products of the intraday return vectors

$$\text{RCov}_t = \sum_{i=1}^n r_{t,i} r'_{t,i} \rightarrow \int_0^1 \Sigma_s ds$$

- ▶ $r_{t,i}$: $N \times 1$ vector of intraday returns; Σ_s : spot covariance matrix; the limit adds the co-jumps (simultaneous jumps); with synchronous, noise-free prices it inherits the CLT of RV
- Asynchronous trading: previous-tick prices create zero returns for the asset that has not traded; covariances shrink towards zero as the interval falls (the Epps effect)
 - ▶ refresh-time sampling (all assets have traded) and the multivariate realised kernel [Barndorff-Nielsen et al. \(2011\)](#): consistent and positive semi-definite
- Forecasting: HAR on the elements of the Cholesky factor of RCov_t keeps forecasts positive definite [Chiriac and Voev \(2011\)](#); HEAVY and Realized GARCH have multivariate versions
- Evaluating covariance forecasts H_t against a matrix proxy Σ_t : robust losses [Laurent, Rombouts and Violante \(2012\)](#)
 - ▶ matrix QLIKE $\ln |H_t| + \text{tr}(H_t^{-1} \Sigma_t)$, with $|\cdot|$ the determinant and tr the trace; Frobenius MSE, the sum of squared element-wise errors

Bitcoin and Ether: realised and DCC correlation



- Daily realised correlation from 5-minute returns (5-day means) and the DCC correlation of daily close-to-close (UTC) returns, 2018–2026

Interpreting the correlations

- Average realised correlation 0.83, average DCC correlation 0.83; correlation between the two series 0.26
- DCC uses one observation a day ($\hat{a} = 0.051$, $\hat{b} = 0.928$): it smooths and reacts with a lag; the realised measure moves daily and drops quickly in idiosyncratic episodes (5% quantile 0.68)
- Combining both (a realised measure in the correlation equation) is the multivariate analogue of Realized GARCH
- Five-minute sampling sits on the flat part of the Epps curve for these two coins (signature slide): the realised correlation is not biased towards zero

Recap: Multivariate

- BEKK guarantees positivity at a quadratic parameter cost; DCC separates variances and correlations and scales to large N
- cDCC fixes the inconsistency of the DCC target; in moderate designs the difference is small
- In high dimension, the target needs shrinkage (DCC-NL); evaluate with GMV risk or robust matrix losses

An open question

- Does exploiting the measurement error of realised variance (HARQ) improve volatility forecasts robustly, or only under particular design choices?
 - ▶ formal: the QLIKE ratio HARQ/HAR and its DM statistic across a pre-registered grid (window, filter, realised measure, asset); falsified if the ratio exceeds 1 or the DM test is insignificant for most cells
 - ▶ the measure of precision (RQ) is itself noisy and fat-tailed, so the gain may be fragile
- Why it matters: risk systems use one-day volatility forecasts every day; a small but robust gain is worth more than a large fragile one
 - ▶ literature to start from: [Bollerslev, Patton and Quaedvlieg \(2016\)](#), [Corsi \(2009\)](#), [Patton \(2011\)](#), [Hansen and Lunde \(2006a\)](#), [Hansen, Lunde and Nason \(2011\)](#)

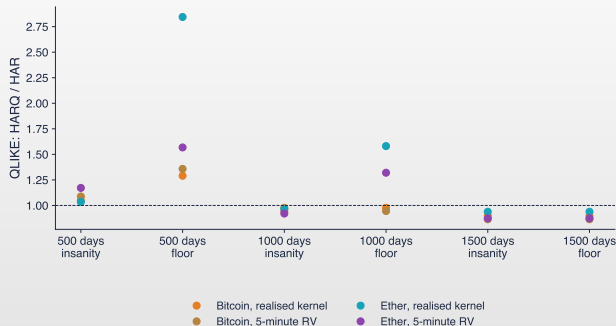
The discovery loop with an AI assistant

- An AI assistant (an LLM such as Claude, ChatGPT, Gemini or Copilot) speeds up each step; Semantic Scholar and Elicit help with the literature
 - ▶ **literature:** List peer-reviewed papers since 2016 that test HARQ-type models out of sample on assets other than US equities; give DOIs. Then check every DOI on Crossref
 - ▶ **hypothesis:** Under which data features (24/7 trading, jumps, zero returns) should the HARQ gain shrink or grow? Give testable predictions.
 - ▶ **code and replication:** ask for HAR and HARQ rolling forecasts, then reproduce a known number first (the Bitcoin QLIKE ratio of this lecture)
 - ▶ **robustness and critique:** Act as a hostile referee: which choices in a HARQ forecast comparison could produce a spurious gain?
- Report: what was asked, what was kept, what was rejected (AI_USE.md, AI_ERRORS.md)

What the human checks

- ▣ Every reference exists and says what is claimed (DOI resolves, title matches, the result is in the paper)
- ▣ The forecast at t uses only data up to t : windows, demeaning of RQ, the insanity filter and the bias correction of log forecasts are all computed in the estimation window
- ▣ The loss is robust (QLIKE, MSE) and the proxy is stated; DM statistics use HAC variances
- ▣ All cells of the grid are reported, not the best one; the number of comparisons is stated
- ▣ Realised measures are recomputed from raw prices and agree with an independent source on overlapping assets and dates

Mini-case: twelve choices, one conclusion?



- QLIKE ratio HARQ/HAR for Bitcoin and Ether: window 500, 1 000 or 1 500 days; insanity filter or a floor at the smallest in-sample RV; target 5-minute RV or realised kernel
- Ratios from 0.865 to 2.844; HARQ better in 58% of the 24 cells, significantly better ($DM < -1.96$) in 50%, significantly worse in 12%: an AI summary reporting one cell would hide most of this

Project idea

- ▣ **Realised volatility of crypto-assets: does the equity toolkit transfer?:** replicate first, then extend
 - ▶ replicate: HAR and HARQ of this lecture for Bitcoin (QLIKE ratio 0.946) and Realized GARCH for the S&P 500 (persistence 0.968)
 - ▶ extend: ten coins from the Binance archive; noise-robust measures from one-second data
 - ▶ extend the tests: an intraday seasonality correction before jump tests; Realized GARCH with the realised kernel
 - ▶ use the spot-ETF date (January 2024) as a natural break
 - ▶ pre-register: assets, sample, measures, models, windows, losses, the DM/MCS tests and the robustness grid
- ▣ Deliverables follow the course rules: repository, report, AI_USE.md, AI_ERRORS.md, oral defence

Key takeaways

- ▣ Gaussian QML needs only the variance equation; with fat tails, use sandwich standard errors
- ▣ Volatility has fast and slow components; macro drivers of the slow one need long samples and honest variance ratios
- ▣ Realised measures make volatility observable with a known error; noise and jumps decide the estimator
- ▣ HAR, HARQ and Realized GARCH beat return-only models one day ahead; evaluate with QLIKE or MSE
- ▣ In many dimensions the correlation target needs shrinkage (DCC-NL); whether the dynamics help must be checked out of sample

Self-assessment

Questions

- ▣ Why are Hessian standard errors too small for GARCH when $\kappa_\eta > 3$?
- ▣ What does the variance ratio of GARCH-MIDAS measure?
- ▣ Why does stable convergence matter for the CLT of RV?
- ▣ Why can a signature plot rise at high frequency?
- ▣ Why is β_{dQ} expected to be negative in HARQ?
- ▣ Which losses keep the ranking under a noisy proxy?

Next:

- ▣ VaR, ES and backtesting: elicibility, scoring and model risk
- ▣ From volatility forecasts to tail forecasts: VaR 1% and ES 2.5%, their scoring functions and backtests

Appendix: the GARCH sandwich

- $\ell_t = -\frac{1}{2}(\ln \sigma_t^2 + \varepsilon_t^2/\sigma_t^2)$, $d_t = \sigma_t^{-2} \partial \sigma_t^2 / \partial \theta$: $s_t = \frac{1}{2}(\eta_t^2 - 1)d_t$ at θ_0
- $B = \mathbb{E} s_t s_t' = \frac{1}{4} \mathbb{E}(\eta_t^2 - 1)^2 \mathbb{E} d_t d_t' = \frac{\kappa_\eta - 1}{4} J$, because η_t is independent of $d_t \in \mathcal{F}_{t-1}$
- $\partial s_t / \partial \theta' = -\frac{1}{2} \eta_t^2 d_t d_t' + \frac{1}{2}(\eta_t^2 - 1) \partial d_t / \partial \theta'$; the second term has mean zero, so $A = \frac{1}{2} J$
- $A^{-1} B A^{-1} = 4J^{-1} \cdot \frac{\kappa_\eta - 1}{4} J \cdot J^{-1} = (\kappa_\eta - 1)J^{-1}$; Hessian only: $A^{-1} = 2J^{-1}$; OPG: $B^{-1} = \frac{4}{\kappa_\eta - 1} J^{-1}$

Appendix: the noise bias of realised variance

- ▣ $\tilde{r}_i = Y_{i/n} - Y_{(i-1)/n} = r_i + u_i - u_{i-1}$, with u i.i.d. $(0, \omega^2)$ independent of X
- ▣ $\sum_i \tilde{r}_i^2 = \sum_i r_i^2 + 2 \sum_i r_i(u_i - u_{i-1}) + \sum_i (u_i - u_{i-1})^2$; the cross term has mean zero, the last has mean $2n\omega^2$
- ▣ $\text{Cov}(\tilde{r}_i, \tilde{r}_{i-1}) = -\omega^2$: first-order negative autocorrelation; the realised kernel adds $2\gamma_1 \approx -2n\omega^2$ back and removes the bias
- ▣ Positively autocorrelated returns (gradual price adjustment, stale prices) give $\gamma_1 > 0$ and a signature plot that rises with the interval; the kernel corrects both signs

Appendix: attenuation and HARQ

- ▣ $IV_{t+1} = c + \phi IV_t + v_{t+1}$, observed $RV_t = IV_t + e_t$, e_t uncorrelated with IV_t and v_{t+1}
- ▣ OLS of RV_{t+1} on RV_t : $\text{plim } \hat{\phi} = \frac{\text{Cov}(IV_{t+1}, IV_t)}{\text{Var}(IV_t) + \text{Var}(e_t)} = \phi\lambda$, $\lambda = \frac{\text{Var}(IV)}{\text{Var}(IV) + \mathbb{E}\text{Var}(e_t)}$
- ▣ The best linear predictor given today's precision uses $\phi\lambda_t$, $\lambda_t = \frac{\text{Var}(IV)}{\text{Var}(IV) + 2IQ_t/n}$: decreasing in IQ_t
- ▣ A first-order expansion in $\sqrt{IQ_t}$ gives $\beta_d + \beta_{dQ}\sqrt{RQ_t}$ with $\beta_{dQ} < 0$: the HARQ specification [Bollerslev, Patton and Quaadvlieg \(2016\)](#)

References (1/5)

- ▣ Aielli, G. P. (2013). Dynamic conditional correlation: On properties and estimation. *Journal of Business & Economic Statistics*, 31(3), 282–299. doi:10.1080/07350015.2013.771027
- ▣ Andersen, T. G., & Bollerslev, T. (1998). Answering the skeptics: Yes, standard volatility models do provide accurate forecasts. *International Economic Review*, 39(4), 885–905. doi:10.2307/2527343
- ▣ Andersen, T. G., Bollerslev, T., & Diebold, F. X. (2007). Roughing it up: Including jump components in the measurement, modeling, and forecasting of return volatility. *The Review of Economics and Statistics*, 89(4), 701–720. doi:10.1162/rest.89.4.701
- ▣ Andersen, T. G., Bollerslev, T., Diebold, F. X., & Labys, P. (2003). Modeling and forecasting realized volatility. *Econometrica*, 71(2), 579–625. doi:10.1111/1468-0262.00418
- ▣ Andersen, T. G., Dobrev, D., & Schaumburg, E. (2012). Jump-robust volatility estimation using nearest neighbor truncation. *Journal of Econometrics*, 169(1), 75–93. doi:10.1016/j.jeconom.2012.01.011
- ▣ Bandi, F. M., & Russell, J. R. (2008). Microstructure noise, realized variance, and optimal sampling. *The Review of Economic Studies*, 75(2), 339–369. doi:10.1111/j.1467-937X.2008.00474.x
- ▣ Barndorff-Nielsen, O. E., Hansen, P. R., Lunde, A., & Shephard, N. (2008). Designing realized kernels to measure the ex post variation of equity prices in the presence of noise. *Econometrica*, 76(6), 1481–1536. doi:10.3982/ECTA6495
- ▣ Barndorff-Nielsen, O. E., Hansen, P. R., Lunde, A., & Shephard, N. (2009). Realized kernels in practice: Trades and quotes. *The Econometrics Journal*, 12(3), C1–C32. doi:10.1111/j.1368-423X.2008.00275.x
- ▣ Barndorff-Nielsen, O. E., Hansen, P. R., Lunde, A., & Shephard, N. (2011). Multivariate realised kernels: Consistent positive semi-definite estimators of the covariation of equity prices with noise and non-synchronous trading. *Journal of Econometrics*, 162(2), 149–169. doi:10.1016/j.jeconom.2010.07.009
- ▣ Barndorff-Nielsen, O. E., & Shephard, N. (2002). Econometric analysis of realized volatility and its use in estimating stochastic volatility models. *Journal of the Royal Statistical Society: Series B*, 64(2), 253–280. doi:10.1111/1467-9868.00336
- ▣ Barndorff-Nielsen, O. E., & Shephard, N. (2004). Power and bipower variation with stochastic volatility and jumps. *Journal of Financial Econometrics*, 2(1), 1–37. doi:10.1093/jjfinec/nbh001
- ▣ Barndorff-Nielsen, O. E., & Shephard, N. (2006). Econometrics of testing for jumps in financial economics using bipower variation. *Journal of Financial Econometrics*, 4(1), 1–30. doi:10.1093/jjfinec/nbi022

References (2/5)

- ▣ Berkes, I., Horváth, L., & Kokoszka, P. (2003). GARCH processes: Structure and estimation. *Bernoulli*, 9(2), 201–227. doi:10.3150/bj/1068128975
- ▣ Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327. doi:10.1016/0304-4076(86)90063-1
- ▣ Bollerslev, T. (1990). Modelling the coherence in short-run nominal exchange rates: A multivariate generalized ARCH model. *The Review of Economics and Statistics*, 72(3), 498–505. doi:10.2307/2109358
- ▣ Bollerslev, T., Patton, A. J., & Quaedvlieg, R. (2016). Exploiting the errors: A simple approach for improved volatility forecasting. *Journal of Econometrics*, 192(1), 1–18. doi:10.1016/j.jeconom.2015.10.007
- ▣ Bollerslev, T., & Wooldridge, J. M. (1992). Quasi-maximum likelihood estimation and inference in dynamic models with time-varying covariances. *Econometric Reviews*, 11(2), 143–172. doi:10.1080/07474939208800229
- ▣ Chiriac, R., & Voev, V. (2011). Modelling and forecasting multivariate realized volatility. *Journal of Applied Econometrics*, 26(6), 922–947. doi:10.1002/jae.1152
- ▣ Comte, F., & Lieberman, O. (2003). Asymptotic theory for multivariate GARCH processes. *Journal of Multivariate Analysis*, 84(1), 61–84. doi:10.1016/S0047-259X(02)00009-X
- ▣ Conrad, C., & Kleen, O. (2020). Two are better than one: Volatility forecasting using multiplicative component GARCH-MIDAS models. *Journal of Applied Econometrics*, 35(1), 19–45. doi:10.1002/jae.2742
- ▣ Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2), 174–196. doi:10.1093/jjfinec/nbp001
- ▣ Diebold, F. X., & Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263. doi:10.1080/07350015.1995.10524599
- ▣ Engle, R. (2002). Dynamic conditional correlation: A simple class of multivariate generalized autoregressive conditional heteroskedasticity models. *Journal of Business & Economic Statistics*, 20(3), 339–350. doi:10.1198/073500102288618487
- ▣ Engle, R. F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4), 987–1007. doi:10.2307/1912773

References (3/5)

- Engle, R. F., Ghysels, E., & Sohn, B. (2013). Stock market volatility and macroeconomic fundamentals. *The Review of Economics and Statistics*, 95(3), 776–797. doi:10.1162/REST_a_00300
- Engle, R. F., & Kroner, K. F. (1995). Multivariate simultaneous generalized ARCH. *Econometric Theory*, 11(1), 122–150. doi:10.1017/S0266466600009063
- Engle, R. F., Ledoit, O., & Wolf, M. (2019). Large dynamic covariance matrices. *Journal of Business & Economic Statistics*, 37(2), 363–375. doi:10.1080/07350015.2017.1345683
- Engle, R. F., & Lee, G. G. J. (1999). A long-run and short-run component model of stock return volatility. In R. F. Engle & H. White (Eds.), *Cointegration, Causality, and Forecasting: A Festschrift in Honour of Clive W. J. Granger* (pp. 475–497). Oxford University Press. doi:10.1093/oso/9780198296836.003.0020
- Engle, R. F., & Sheppard, K. (2001). Theoretical and empirical properties of dynamic conditional correlation multivariate GARCH. *NBER Working Paper* 8554. doi:10.3386/w8554
- Epps, T. W. (1979). Comovements in stock prices in the very short run. *Journal of the American Statistical Association*, 74(366), 291–298. doi:10.1080/01621459.1979.10482508
- Francq, C., & Zakoian, J.-M. (2004). Maximum likelihood estimation of pure GARCH and ARMA-GARCH processes. *Bernoulli*, 10(4), 605–637. doi:10.3150/bj/1093265632
- Francq, C., & Zakoian, J.-M. (2019). *GARCH Models: Structure, Statistical Inference and Financial Applications* (2nd ed.). Wiley. doi:10.1002/9781119313472
- Ghysels, E., Santa-Clara, P., & Valkanov, R. (2006). Predicting volatility: Getting the most out of return data sampled at different frequencies. *Journal of Econometrics*, 131(1–2), 59–95. doi:10.1016/j.jeconom.2005.01.004
- Glosten, L. R., Jagannathan, R., & Runkle, D. E. (1993). On the relation between the expected value and the volatility of the nominal excess return on stocks. *The Journal of Finance*, 48(5), 1779–1801. doi:10.1111/j.1540-6261.1993.tb05128.x
- Hafner, C. M., & Preminger, A. (2009). On asymptotic theory for multivariate GARCH models. *Journal of Multivariate Analysis*, 100(9), 2044–2054. doi:10.1016/j.jmva.2009.03.011
- Hamilton, J. D. (1994). *Time Series Analysis*. Princeton University Press. doi:10.2307/j.ctv14jx6sm

References (4/5)

- Hansen, P. R., & Huang, Z. (2016). Exponential GARCH modeling with realized measures of volatility. *Journal of Business & Economic Statistics*, 34(2), 269–287. doi:10.1080/07350015.2015.1038543
- Hansen, P. R., Huang, Z., & Shek, H. H. (2012). Realized GARCH: A joint model for returns and realized measures of volatility. *Journal of Applied Econometrics*, 27(6), 877–906. doi:10.1002/jae.1234
- Hansen, P. R., & Lunde, A. (2005). A forecast comparison of volatility models: Does anything beat a GARCH(1,1)? *Journal of Applied Econometrics*, 20(7), 873–889. doi:10.1002/jae.800
- Hansen, P. R., & Lunde, A. (2006a). Consistent ranking of volatility models. *Journal of Econometrics*, 131(1–2), 97–121. doi:10.1016/j.jeconom.2005.01.005
- Hansen, P. R., & Lunde, A. (2006b). Realized variance and market microstructure noise. *Journal of Business & Economic Statistics*, 24(2), 127–161. doi:10.1198/073500106000000071
- Hansen, P. R., Lunde, A., & Nason, J. M. (2011). The model confidence set. *Econometrica*, 79(2), 453–497. doi:10.3982/ECTA5771
- Heber, G., Lunde, A., Shephard, N., & Sheppard, K. (2009). *Oxford-Man Institute's realized library*, version 0.3. Oxford-Man Institute, University of Oxford (discontinued in 2022; copy of 28 February 2022 preserved by the Internet Archive). web.archive.org
- Huang, X., & Tauchen, G. (2005). The relative contribution of jumps to total price variance. *Journal of Financial Econometrics*, 3(4), 456–499. doi:10.1093/jjfinec/nbi025
- Jacod, J., Li, Y., Mykland, P. A., Podolskij, M., & Vetter, M. (2009). Microstructure noise in the continuous case: The pre-averaging approach. *Stochastic Processes and their Applications*, 119(7), 2249–2276. doi:10.1016/j.spa.2008.11.004
- Jensen, S. T., & Rahbek, A. (2004). Asymptotic inference for nonstationary GARCH. *Econometric Theory*, 20(6), 1203–1226. doi:10.1017/S0266466604206065
- Laurent, S., Rombouts, J. V. K., & Violante, F. (2012). On the forecasting accuracy of multivariate GARCH models. *Journal of Applied Econometrics*, 27(6), 934–955. doi:10.1002/jae.1248
- Ledoit, O., & Wolf, M. (2004). A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2), 365–411. doi:10.1016/S0047-259X(03)00096-4

References (5/5)

- ❑ Ledoit, O., & Wolf, M. (2020). Analytical nonlinear shrinkage of large-dimensional covariance matrices. *The Annals of Statistics*, 48(5), 3043–3065. doi:10.1214/19-AOS1921
- ❑ Lee, S. S., & Mykland, P. A. (2008). Jumps in financial markets: A new nonparametric test and jump dynamics. *The Review of Financial Studies*, 21(6), 2535–2563. doi:10.1093/rfs/hhm056
- ❑ Lee, S.-W., & Hansen, B. E. (1994). Asymptotic theory for the GARCH(1,1) quasi-maximum likelihood estimator. *Econometric Theory*, 10(1), 29–52. doi:10.1017/S0266466600008215
- ❑ Ling, S., & McAleer, M. (2003). Asymptotic theory for a vector ARMA-GARCH model. *Econometric Theory*, 19(2), 280–310. doi:10.1017/S0266466603192092
- ❑ Lumsdaine, R. L. (1996). Consistency and asymptotic normality of the quasi-maximum likelihood estimator in IGARCH(1,1) and covariance stationary GARCH(1,1) models. *Econometrica*, 64(3), 575–596. doi:10.2307/2171862
- ❑ Newey, W. K., & West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708. doi:10.2307/1913610
- ❑ Pakel, C., Shephard, N., Sheppard, K., & Engle, R. F. (2021). Fitting vast dimensional time-varying covariance models. *Journal of Business & Economic Statistics*, 39(3), 652–668. doi:10.1080/07350015.2020.1713795
- ❑ Patton, A. J. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1), 246–256. doi:10.1016/j.jeconom.2010.03.034
- ❑ Patton, A. J., & Sheppard, K. (2015). Good volatility, bad volatility: Signed jumps and the persistence of volatility. *The Review of Economics and Statistics*, 97(3), 683–697. doi:10.1162/REST_a_00503
- ❑ Petropoulos, F., Apiletti, D., Assimakopoulos, V., Babai, M. Z., et al. (2022). Forecasting: Theory and practice. *International Journal of Forecasting*, 38(3), 705–871. doi:10.1016/j.ijforecast.2021.11.001
- ❑ Shephard, N., & Sheppard, K. (2010). Realising the future: Forecasting with high-frequency-based volatility (HEAVY) models. *Journal of Applied Econometrics*, 25(2), 197–231. doi:10.1002/jae.1158
- ❑ Zhang, L., Mykland, P. A., & Aït-Sahalia, Y. (2005). A tale of two time scales: Determining integrated volatility with noisy high-frequency data. *Journal of the American Statistical Association*, 100(472), 1394–1411. doi:10.1198/016214505000000169